Formalizing the Reduced Mahler $3/2$ Dynamical System in Lean 4: Exact 1D Slack Reduction, Constructive Coasting Survivors, Conditional Finiteness Theorems, and Certified Transcendence Barriers

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Toolchain: Lean 4.19.0 + Mathlib v4.19.0 (`142` `.lean` files, `0` `sorry`s, `0` warnings, standard Lean axioms `[propext, Classical.choice, Quot.sound]`)

Abstract

Mahler's $3/2$ problem (1968) asks whether there exists any real "Z-number" $\alpha > 0$ such that the fractional part $\{(3/2)^n \alpha\}$ lies in $[0, 1/2)$ for every integer $n \geq 0$. Writing $\alpha = a_0 + \xi$ with $a_0 = \lfloor \alpha \rfloor \in \mathbb{N}$ and $\xi \in [0, 1)$, survival up to step N is equivalent to the discrete ceiling recurrence $g_{n+1} = \lceil 3g_n/2 \rceil = \lfloor (3g_n + 1)/2 \rfloor$ ($g_0 = a_0$) admitting a non-empty intersection of N nested archimedean companion intervals $[\mathrm{lowerBound}(a_0, n),\ \mathrm{upperBound}(a_0, n))$ (`InfiniteZNumberBridgeAndRationalBase.lean`).

In this work, we present a 142-file, `sorry`-free formalization in Lean 4 that investigates both **constructive survival** and **obstructions to infinite orbits**: 1. **Exact 1D Slack Reduction & Infinite-Horizon Z-Number Bridge:** By grouping each odd step $g \mapsto (3g + 1)/2$ with its $s(k) - 1$ subsequent even steps $g \mapsto 3g/2$ via $3b_k + 1 = 2^{s(k)} u(k)$ ($\gcd(u(k), 6) = 1$), we collapse all $\binom{K}{2}$ pairwise companion-interval inequalities along a finite block chain into a **single 1D recurrence** (`minSlack`), and prove its exact two-way equivalence with the existence of a real companion surviving through step $T(K)$ (both in interval form and in direct `Int.fract` form). Normalizing by $2^{T(m)}$ yields the time-homogeneous 1D slack map $\mu_2(s, m+1) = \min\left(1,\ (3/2)^{s(m)} \mu_2(s, m) - 2\right)$ driven by the arithmetic-controlled block length $s(m) = v_2(3b_m + 1)$ (`CoupledFractionalPartDynamics.lean` , `FiniteSeedCarryAndBackwardDescent.lean`). 2. **Constructive Survivors & 113-Step Record-Holder:** We prove that any block sequence with $s(k) \geq 3$ saturates the ceiling minSlack $s\ m = 2^{T(m)}$ and automatically admits a valid real companion at every finite horizon. Using this sufficiency theorem, we construct closed-form seed families surviving arbitrarily many blocks prior to ceiling exit, and machine-certify a 55-bit integer (`recordHolderA0 = 29871555608238677 < 2^55`) that survives **113 steps** (58 steps past its 55-bit binary length) before dying at step 114 (`MahlerGrandSynthesis.lean` , `InfiniteZNumberBridgeAndRationalBase.lean`). 3. **Unconditional Exclusion of Restricted, Aperiodic, Non-Stationary & Rapid-Growth Classes:** We prove explicit, logarithm-free step cutoffs in Lean 4 ruling out: (i) sub-ceiling bounded-block runs ($s(k) \leq c$), (ii) periodic and uniform-macro-block sequences (`2^(D m) | ((3^D - 2^D)(3a_0+1) + 2^(s 0) R_0)`), (iii) **arbitrary two-letter $\{a, b\}$ ($3 \leq a \leq b$) mechanical/Sturmian block chains** (`no_infinite_mechanical_two_letter_block_chain` , with factor complexity $\leq L + 1$ derived directly from phase thresholds), (iv) **sub-critical liminf-complexity sequences** ($\kappa < \log 2 / \log(3/2) \approx 1.709511$) under a positive Cesàro mean, uniform window balance, or scale-local epoch balance (`no_infinite_tendsto_mean_log_threshold_block_chain` , `no_infinite_uniformWindowMean_global_complexity_block_chain` , `no_infinite_scale_local_balanced_block_chain` , including the non-stationary oscillating-epoch family `nonStationaryEpochSeq a b`), (v) **3-step encoded Thue–Morse and Period-Doubling parity streams** (`no_nonneg_seed_realizes_tmEncodedParity` , `pdEncodedParity_cutoff_and_exclusion` , with quantitative seed-residue lower bounds $\geq 2^{2^k}$) and **Thue–Morse block-length chains** (`no_infinite_tmBlockSeq_block_chain`), and (vi) **sustained rapid-growth chains** ($3 \leq s(m+1)$ and $3^{s(m)} < 2^{s(m+1)}$, via well-founded odd-quotient descent $u(m+1) + 2 \leq u(m)$ in `no_infinite_rapid_growth_block_chain`). We also formalize the exact arithmetic setup for Baker–Matveev, Ridout, and Schlickewei Subspace analyses, clearly separating the machine-checked algebraic/analytic bounds from external transcendence theorems not yet in Mathlib. 4. **Exact Coupled Seed-Residue State, 5-Way General Conjecture Equivalence, Certified $B(29) \geq 4096$ Bound & Structural Barriers:** We formalize the denominator-cleared integer feasibility margin $D_{N+1} = \min(2^{N+1},\ 3D_N - 2^{N+1}(\varepsilon_N \bmod 2))$, the coupled quotient–residue–margin transition $2q_{N+1} = 3q_N + 3^{N+1}\eta_N + e_N$, the **5-way general Mahler conjecture equivalence** (`general_mahler_conjecture_iff_positive_stabilized_branch_escape` , `mahler_conjecture_iff_leastPositiveSurvivor_unbounded`), the **Lean-certified global step-29 lower bound** $m \geq 4096$ **($B(29) \geq 4096$) and extremal seed** `149` (`positive_survivor_step29_ge_4096` , `seed_149_survives_28_and_fails_29`), and the exact ceiling-recovery theorem (`signedMargin_100_ceiling_recovery`) explaining why $s(m) \geq 3$ (`100`) blocks escape pure scalar-margin contraction and why the general conjecture remains open.

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1. Introduction: From Informal Proof Brainstorming to Machine-Checked Rigor

When attempting to prove Mahler's $3/2$ conjecture or related Collatz-type problems, informal proof sketches typically fall into two traps: 1. **The Probabilistic / Density Fallacy:** Treating 2-adic parity constraints and archimedean interval constraints as statistically independent, or assuming that once the step count T exceeds the bit-length $W = \lfloor \log_2 a_0 \rfloor + 1$ (the "coasting regime"), a trajectory can only survive a handful of steps. 2. **The Hidden Rank-Collapse Trap:** Constructing auxiliary linear combinations, Wronskians, or multi-origin determinants that appear to overdetermine the orbit, without noticing that the underlying 1D affine recurrence forces those combinations to be algebraically dependent.

By formalizing every proposed attack vector in Lean 4—and requiring that every failed approach be diagnosed via a machine-checked counterexample or algebraic identity—we separated genuine number-theoretic structure from mirages.

1.5 Literature Survey, Attribution of Prior Work & Novel Contributions

To place this formalization in its proper mathematical context, we carefully distinguish **prior theorems from the literature that we rediscovered or formalized** from **novel formulations, theorems, and certified barrier analyses originating in this work**.

A. Attribution to Prior Mathematical Literature

1. **Mahler (1968) — Discrete Companion Reduction, Series Formula & Two-Seed Ridout Bound [1]:**
2. Kurt Mahler introduced Z-numbers and proved the foundational reduction: for each integer $g_0 = a_0 \geq 1$, there is *at most one* real companion $\alpha = a_0 + \xi$ ($\xi \in [0,1)$) such that α is a Z-number, governed by the discrete ceiling recurrence $g_{n+1} = \lceil 3g_n/2 \rceil = \lfloor (3g_n+1)/2 \rfloor$ and nested archimedean intervals (`MahlerDemandA.lean` , `PureIntegerPinch.lean` , `InfiniteZNumberBridgeAndRationalBase.lean`). As we formally verify in `odd_step_floor_exceeds_half_and_forces_ceil` , when g_n is odd and $0 \leq r_n < 1/2$, the remainder relative to $\lfloor 3g_n/2 \rfloor$ is $1/2 + 3r_n/2 \geq 1/2$, so keeping $\{(3/2)^{n+1}\alpha\} < 1/2$ forces $r_n \geq 1/3$ and $\lfloor (3/2)^{n+1}\alpha \rfloor = \lceil 3g_n/2 \rceil$.
3. Mahler also derived the infinite series representation of the unique candidate companion ξ_∞ (`InfiniteSeries.lean` , `TelescopingSeriesDuality.lean`) and applied **Ridout's Theorem (1957) [2]** to the difference of two hypothetical Z-numbers in the same binary interval $[2^k, 2^{k+1})$ to show that at most finitely many Z-numbers exist per interval (`mahler_two_seed_same_step_ridout_bound` and `two_seed_shared_prefix_divisibility` in `RidoutSubspaceFramework.lean`), yielding $|\{Z\text{-numbers} \leq X\}| = O(X^{0.7})$.
4. **Vijayaraghavan (1940), Pollington (1979) & Choquet (1980) — Distribution of Powers of $3/2$ and the $1/3$ Range Gap [3, 4, 5]:**
5. Vijayaraghavan [3] proved that fractional parts of rational powers $(p/q)^n\alpha$ for $\alpha \in \mathbb{Q}^\times$ have infinitely many limit points and cannot stay confined to $[0, 1/2)$.
6. Pollington [4] and Choquet [5] established lower bounds on the oscillation of $\{(3/2)^n\alpha\}$, leading to the critical gap constant $1/3$.
7. **Flatto, Lagarias & Pollington (1995) — The $1/3$ Range Theorem, Symbolic Dynamics & Forbidden Subwords [6]:**
8. In their landmark paper [6], Flatto, Lagarias, and Pollington proved that for every real $\alpha > 0$,

$$\limsup_{n\to\infty} \{(3/2)^n\alpha\} - \liminf_{n\to\infty} \{(3/2)^n\alpha\} \geq \frac{3-2}{3} = \frac{1}{3}.$$

   This $1/3$ range gap is the continuous foundation of our $\frac{1}{3}$ **Archimedean Drain Budget** ($\sum_{k=1}^{m} (2/3)^{T(k)+1} < 1/3$ in `TelescopingSeriesDuality.lean`) and explains why $s(k) = 1$ (which consumes $(2/3)^2 = 4/9 > 1/3$ in one block) is forbidden (`EscapeBranchTwoCap.lean`).
9. Flatto, Lagarias, and Pollington also developed the **symbolic dynamics of the parity sequence** $g_n \bmod 2$, characterizing admissible finite binary words via an infinite hierarchy of forbidden subwords and deriving average step-rate / topological entropy bounds. Our theorems in `GlobalStepRateBounds.lean` and `MinSlackDynamicalSystem.lean` —including $s(k) \geq 2$, the post-2 recovery requirements $s(k+1) \geq 6, s(k+2) \geq 3, s(k+3) \geq 3$, the $(2, 6, 3, 3, 2, 10, 7)$ cascade, and the $3m - 1$ step-rate bound—rediscover and machine-check these symbolic-dynamical constraints in odd-block length coordinates $s(k)$.
10. **Classical Diophantine & Transcendence Machinery (Ridout [2], Schlickewei [7], Stewart [8], Baker–Wüstholz / Matveev [9, 10], Corvaja–Zannier [11, 12], Adamczewski–Bugeaud [13], Dubickas [14]) & Rational-Base Numeration Systems (Akiyama–Frougny–Sakarovitch [16], Andrieu–Eliahou–Vivion [17]):**
11. Our analyses in Modules 89–102 formalize the exact algebraic and logarithmic inequalities feeding into the classical frameworks of linear forms in logarithms (Baker–Wüstholz [9], Matveev [10]), p-adic Diophantine approximation (Ridout [2], Schlickewei's p-adic Subspace Theorem [7]), simultaneous representation in bases 2 and 3 (Stewart [8]), S-unit GCD bounds (Corvaja–Zannier [11, 12]), and subword-complexity transcendence criteria (Adamczewski–Bugeaud [13]). Because the deep transcendence theorems of Matveev, Ridout, and Schlickewei are not themselves formalized in Mathlib, our Lean modules establish the unconditional algebraic/archimedean upper bounds, GCD divisibility laws, and hyperplane syzygies directly, while clearly noting where external transcendence theorems are invoked in conditional interpretations.
12. Akiyama, Frougny, and Sakarovitch (2008) [16] introduced rational-base number systems $\mathrm{rep}_{p/q}(n)$, and Andrieu, Eliahou, and Vivion (2025) [17] proved that Borel normality of minimal/maximal rational-base words (or residue equidistribution of $T_{3/2}^n(a_0) = \lceil 3x/2 \rceil \bmod 2^k$) implies the non-existence of Z-numbers. In `InfiniteZNumberBridgeAndRationalBase.lean` , we connect `Mahler.step` directly to $T_{3/2}$ and prove that single deterministic occurrences of forbidden patterns suffice without assuming full normality.

B. Novel Contributions Originating in This Work (and Comparison with Prior Lean Formalizations)

1. **Relationship to Prior Lean Repositories on Mahler's Work:**
2. A direct search of all public Lean repositories on GitHub shows that **Google DeepMind's `formal-conjectures` project [15]** (`FormalConjectures/Wikipedia/Mahler32.lean` , `FormalConjectures/Books/BugeaudDistributionModuloOne/Problem10_9.lean` , and `FormalConjecturesForMathlib/NumberTheory/DiophantineApproximation/ZNumber.lean` , 2025–2026) previously formalized the *definition* `IsZNumber` , a 5-line sanity check `¬ IsZNumber 1` , and the *statements* (with `sorry` placeholders) of Mahler's conjecture, the consequence of $\Omega(3/2) > 1/2$, the Flatto–Lagarias–Pollington bound $\Omega(p/q) \geq 1/p$, Lebesgue measure zero, and Hausdorff dimension < 1.
3. Other Lean 4 repositories bearing Mahler's name address distinct theorems of Kurt Mahler: `giuliocaflisch/mahler` (Mahler's theorem on p-adic continuous functions), `DiegoMarquesMath/MahlerLean` (Mahler's classification of Liouville numbers), and `baobingzhang/jsp-000025-lehmer-schinzel-lean` (polynomial Mahler measure).
4. Thus, while `google-deepmind/formal-conjectures` [15] formalized the top-level conjecture statements for Z-numbers, our 122-module `sorry` -free library is a substantial proved Lean 4 development of the discrete Mahler $3/2$ dynamical system, the finite and infinite companion-interval reductions, the Flatto–Lagarias–Pollington symbolic forbidden-subword constraints in odd-block coordinates, the Sturmian/balanced aperiodic complexity exclusions, the finite-seed vs. 2-adic isolation theorems, and the Diophantine setup identities.
5. **The Reduced Coprime Quotient System & Exact 1D `minSlack` Dynamical Collapse (`CoprimeRemainderSlack.lean` , `MinSlackDynamicalSystem.lean` , `ConverseCompanionReconstruction.lean` , `CoupledFractionalPartDynamics.lean`):**

6. Formulating the dynamics in terms of the coprime odd quotients $2^{s(k+1)}u(k+1) = 3^{s(k)}u(k) + 1$ ($\gcd(u(k), 6) = 1$), the 3-adic block-boundary factorization $b_{k+1} = 3^{s(k)-1}u(k)$, and the polynomial $R(s, m)$.
7. Collapsing all $\binom{K}{2}$ shifted archimedean companion inequalities into the **single 1D recurrence** `minSlack s (m+1) = min (2^(T(m+1))) (3^(s m) * minSlack s m - 2^(T(m+1)+1))` and its normalized 1D update $\mu_2(s, m+1) = \min\left(1,\ (3/2)^{s(m)}\mu_2(s, m) - 2\right)$ driven by the block length $s(m)$ (with uncapped values in $(0, 1]$ partitioned by $(0, 1] = \bigsqcup_{c=2}^{\infty}(2(2/3)^c, 3(2/3)^c] \cap (0, 1]$), together with the two-way **Converse Companion Reconstruction Theorem** (`(∃ ξ, ValidCompanionUpTo a_0 ξ (blockSteps s K)) ↔ 0 < minSlack s (K - 1)`).
8. **Additive Separation of Shifted Slacks & the Three-Origin Subspace Syzygy** (`MultiOriginSubspaceSyzygy.lean`):
9. Proving that the two-parameter triangle of shifted slacks separates additively as $\frac{2^{T(r)}\Delta_{m-r}^{(r)}}{3^{T(m)}} = \frac{\Delta_m^{(0)}}{3^{T(m)}} - \frac{\Delta_r^{(0)} - 2^{T(r)}}{3^{T(r)}}$, eliminating `shiftSeq` completely from `0 < minSlack s m`, and proving the exact **Three-Origin Schlickewei Subspace Hyperplane Syzygy** (Dead End 15) showing that any ≥ 3 shifted origins lie in the 2D plane $\mathrm{span}\{\Delta_m^{(0)}, 3^{T(m)}\}$.
10. **Carryless Base-3 Frozen-Prefix Equivalence & Dual Remainder Complementarity** (`SimultaneousBaseDigitRigidity.lean`):
11. Proving $0 < \Delta_m \iff \lfloor 2^{T(m+1)+1}u(m)/3^{T(m)} \rfloor = 6a_0 + 2$ (and its multi-origin companion equivalence) together with the exact dual remainder law $(P_m \bmod 3^{T(m)}) + (Q_m \bmod 2^{T(m)}) = 3^{T(m)}$.
12. **Exact Three-Way Multi-Step GCD Identities & Two-Sided CRT Quotient Floors** (`MultiStepQuotientGCD.lean`, `ThreeAdicAndSeedGCDLocks.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `Mod216TwoSidedQuotientFloor.lean`):
13. Proving the exact GCD equalities $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)}) = \gcd(u(r+d), R_d^{(r)})$, $\gcd(\Delta_r, \Delta_{r+d}) = \gcd(\Delta_r, R_d^{(r)})$ (with $\gcd(\Delta_r, \Delta_{r+1}) = 1$), and $\gcd(u(m), \Delta_m) = \gcd(u(m), 2a_0 + 1) = \gcd(\Delta_m, 2a_0 + 1)$.
14. Proving the closed-form 3-adic residue lock $u(k+1) \bmod 3^{s(k)} = ((3^{s(k)} + 1)/2)^{s(k+1)} \bmod 3^{s(k)}$ and combining backward 3-adic congruences ($\bmod\ 3, 9, 27$) with forward 2-adic congruences ($\bmod\ 4, 8, 64, 256$) to prove $u(k) \neq 1$ ($k \geq 1$), $1735 \leq u(k)$ at $s(k) = 2$, $65 \leq u(k)$ across $2 \leq s(k) \leq 6$, and $103 \leq u(k)$ on ceiling-saturated runs.
15. **Exact $2^{T(m)}$-Cancellation in the Single-Origin Ridout Product & Explicit Repeated-Subword Cutoffs** (`RidoutSubspaceFramework.lean`, `RepeatedSubwordComplexity.lean`):
16. Proving the single-origin cancellation $2^{s(m)+D+1}u(m)\Delta_m < 3^{T(m)+1}(2a_0 + 1)$ and the exponent identity $(1 - \frac{\log 2}{\log 3}) + \frac{\log 2}{\log 3} + 1 = 2$ (Dead End 14).
17. Proving the parity-enhanced common multiplier $u(r_2) - u(r_1) = 2^{S_{\text{tail}}+1}K_0$, $u(r_2 + d) - u(r_1 + d) = 2 \cdot 3^S K_0$ across any repeated subword and its logarithm-free linear cutoff $\text{blockSteps}^{(r_1)}L \leq T(r_2) + W$.
18. **Discovery & Machine Certification of the 55-Bit, 113-Step Coasting Record-Holder** (`MahlerGrandSynthesis.lean`):
19. Using our ceiling-saturation sufficiency theorem (`exists_validCompanionUpTo_of_three_le`) to construct a targeted congruence search that uncovered `recordHolderA0 = 29871555608238677 < 2^55`, surviving 113 steps (58 steps past its 55-bit binary length) and dying at step 114.

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2. The Odd-Block Reduction & Dual 2-Adic / 3-Adic Recurrence

Let `step x = (3 * x + 1) / 2` if `x` is odd and `3 * x / 2` if `x` is even, and let `orbit a_0 t` be the t-th iterate starting from $a_0 \in \mathbb{N}$.

A sequence of block lengths $s : \mathbb{N} \to \mathbb{N}$ and odd quotients $u : \mathbb{N} \to \mathbb{N}$ forms a **`ValidBlockChain a_0 s u K`** (`OddBlockReduction.lean`) if for every $k < K$:

$$1 \leq s(k), \qquad u(k) \bmod 2 = 1, \qquad 3 \cdot b_k + 1 = 2^{s(k)}u(k),$$

where $b_k :=$ orbit a_0 $(T(k))$ and $T(k) :=$ blockSteps s $k = \sum_{i=0}^{k-1} s(i)$.

Fundamental Identities Along Any Valid Block Chain

- **One-Step Coprime Quotient Recurrence** (`CoprimeRemainderSlack.lean`):

$$2^{s(k+1)}u(k+1) = 3^{s(k)}u(k) + 1 \qquad (k + 1 < K).$$

- **Exact 3-Adic Factorization of Block-Boundary Orbit Values** (`ThreeAdicAndSeedGCDLocks.lean`):

$$b_{k+1} = \text{orbit } a_0\ (T(k+1)) = 3^{s(k)-1} \cdot u(k), \qquad \gcd(u(k), 6) = 1.$$

  Thus at every block boundary b_{k+1}, its **3-adic valuation** is $s(k) - 1$ (encoding the *preceding* block length) with 3-free part $u(k)$, while the **2-adic valuation of** $3b_{k+1} + 1$ is $s(k+1)$ (encoding the *upcoming* block length) with 2-free part $u(k+1)$!
- **Multi-Step Coprime Remainder & Intrinsic Slack** (`CoprimeRemainderSlack.lean`, `AperiodicSlackCocycle.lean`): Defining $R(s, 0) = 0$ and $R(s, m+1) = 3^{s(m)}R(s, m) + 2^{\text{tailSteps } s\ m}$ (where tailSteps s $m = \sum_{i=1}^{m} s(i)$), the m-step quotient satisfies:

$$2^{\text{tailSteps } s\ m}u(m) = 3^{T(m)}u(0) + R(s, m).$$

  Eliminating $u(0)$ via $2^{s(0)}u(0) = 3a_0 + 1$ yields the **Orbit Slack Identity**:

$$\Delta_m := \text{integerSlack } s\ m := 3^{T(m)} - 2^{s(0)+1}R(s, m) = 3^{T(m)+1}(2a_0 + 1) - 2^{T(m+1)+1}u(m).$$

  Notice that the middle expression $3^{T(m)} - 2^{s(0)+1}R(s, m)$ depends **solely on the block sequence** s and is completely independent of a_0 and u!

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3. Exact 1D Dynamical Reduction & Converse Companion Reconstruction

At first glance, ensuring that all K archimedean companion intervals $[\text{lowerBound}(a_0, T(j)),\ \text{upperBound}(a_0, T(m+1)))$ overlap requires checking $\binom{K}{2}$ pairwise inequalities:

$$\forall\, 0 \le j \le m < K, \qquad 0 < \Delta^{(j)}_{m-j} := \text{integerSlack (shiftSeq } s\ j)\ (m-j).$$

We proved four equivalent ways to collapse this $\binom{K}{2}$ triangular system into a **1D dynamical condition**:

1. **The 1D `minSlack` Dynamical System (`MinSlackDynamicalSystem.lean`, `ConverseCompanionReconstruction.lean`):** Define:

   $$\text{minSlack } s\ 0 = 1, \qquad \text{minSlack } s\ (m+1) = \min\Big(2^{T(m+1)},\ 3^{s(m)} \cdot \text{minSlack } s\ m - 2^{T(m+1)+1}\Big).$$

   Then for any `ValidBlockChain a_0 s u K` ($K \ge 1$), a real companion $\xi \in \mathbb{R}$ surviving through step $T(K)$ exists **if and only if** `0 < minSlack s (K - 1)`:

   $$\big(\exists \xi : \mathbb{R},\ \text{ValidCompanionUpTo } a_0\ \xi\ (T(K))\big) \iff 0 < \text{minSlack } s\ (K-1).$$

2. **Shift-Free Unshifted Ceiling Comparison (`MultiOriginSubspaceSyzygy.lean`):** Using the **Shifted Slack Affine Collapse Identity**

   $$2^{T(r)}\Delta^{(r)}_d = \Delta^{(0)}_{r+d} + 3^{S_{r,d}}\big(2^{T(r)} - \Delta^{(0)}_r\big),$$

   every `shiftSeq` can be eliminated:

   $$0 < \text{minSlack } s\ m \iff \forall j \le m, \quad 3^{T(m)}\big(\Delta^{(0)}_j - 2^{T(j)}\big) < 3^{T(j)}\Delta^{(0)}_m.$$

3. **Carryless Base-3 Shift-and-Add Automaton (`SimultaneousBaseDigitRigidity.lean`):** The scaled quotients $P_m := 2^{T(m+1)+1}u(m)$ satisfy the base-3 shift-and-add recurrence $P_{m+1} = 3^{s(m)}P_m + 2^{T(m+1)+1}$, and unconditionally $P_m = 3^{T(m)}(6a_0+2) + 2^{s(0)+1}R(s,m)$. Thus:

   $$0 < \Delta^{(0)}_m \iff \left\lfloor \frac{2^{T(m+1)+1}u(m)}{3^{T(m)}} \right\rfloor = 6a_0 + 2 \quad (\text{no base-3 carry across } 3^{T(m)}!).$$

4. **Time-Homogeneous 1D Slack Map & Lüroth–Markov Partition (`CoupledFractionalPartDynamics.lean`, `FiniteSeedCarryAndBackwardDescent.lean`):** Normalizing by $2^{T(m)}$ via $\mu_2(s,m) := \text{minSlack } s\ m / 2^{T(m)}$ removes the explicit time dependence $T(m)$ from the recurrence:

   $$\mu_2(s,0) = 1, \qquad \mu_2(s,m+1) = \min\left(1,\ \left(\frac{3}{2}\right)^{s(m)}\mu_2(s,m) - 2\right).$$

   (Scope note: μ_2 still takes the block length $s(m)$ as input—at $\mu_2 = 1$, every $s(m) \ge 3$ preserves $\mu_2 = 1$. Along an integer orbit, $s(m) = v_2(3b_m+1)$ and $u(m) = (3b_m+1)/2^{s(m)}$ are uniquely forced by the arithmetic boundary b_m, as proved in `arith_block_parameters_unique`.) Whenever the uncapped ratio $\theta_2(s,m) = \Delta_m/2^{T(m)}$ transitions into $(0,1]$, $\theta_2(s,m)$ must lie in the partition cell $I_{s(m)} = (2(2/3)^{s(m)},\ 3(2/3)^{s(m)}]$. Because $3(2/3)^{c+1} = 2(2/3)^c$, the cells $I_c \cap (0,1]$ ($c = 2, 3, 4, \ldots$) form a **pairwise-disjoint partition of** $(0,1]$.

5. **Infinite-Horizon Z-Number Bridge & Direct `Int.fract` Equivalence (`InfiniteZNumberBridgeAndRationalBase.lean`):** For any $a_0 \in \mathbb{N}$ and $\alpha \in \mathbb{R}$, `ValidCompanionUpTo a_0 α N` holds **if and only if** $\lfloor \alpha \rfloor = a_0$ and $\forall n \le N,\ \text{Int.fract}(\alpha(3/2)^n) < 1/2$ (`validCompanionUpTo_iff_floor_and_fract_lt_half`), and `ValidCompanionForever a_0 α` holds **if and only if** $\lfloor \alpha \rfloor = a_0$ and $\forall n \in \mathbb{N},\ \text{Int.fract}(\alpha(3/2)^n) < 1/2$ (`validCompanionForever_iff_floor_and_forall_fract_lt_half`). Along an infinite block chain, any α_∞ satisfying the strict block-boundary separation $\forall m,\ \text{lowerBound } a_0\ (T(m)) \le \alpha_\infty < \text{upperBound } a_0\ (T(m))$ is a valid companion across all steps (`validCompanionForever_of_block_boundary_separation`).

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4. Ceiling Saturation & the 113-Step 55-Bit Record-Holder

Whenever $s(k) \ge 3$ for all $k < m$, `minSlack` remains pinned at its ceiling (`CeilingSaturationSynthesis.lean`):

$$\big(\forall k < m,\ 3 \le s(k)\big) \implies \text{minSlack } s\ m = 2^{T(m)} > 0.$$

Consequently, **every** block chain with $s(k) \ge 3$ automatically satisfies all archimedean companion constraints (`exists_validCompanionUpTo_of_three_le`).

- **Arbitrarily Long Closed-Form Survivors (`ExplicitHorizonCutoffs.lean`):** The closed-form integer family

  $$a_0(r) = \frac{92 \cdot 2^{12r} - 27}{65}$$

  executes r consecutive $(3,3,3,3)$ cycles ($12r$ steps) with $s(k) = 3$, surviving arbitrarily many steps before its first $s = 2$ block.
- **The 113-Step Coasting Record-Holder (`MahlerGrandSynthesis.lean`):** Searching the ceiling-saturated congruence tree uncovered the 55-bit seed:

  $$\text{recordHolderA0} = 29{,}871{,}555{,}608{,}238{,}677 < 2^{55}.$$

In `recordHolderA0_exact_113_survival_and_114_death`, Lean machine-checks that `recordHolderA0` executes 31 blocks with $s(k) \geq 3$ through $T(31) = 109$, hits its first $s(31) = 2$ at step 109, takes $s(32) = 1$ at step 111, **survives 113 steps** (58 steps past its 55-bit binary length!), and dies at step 114.

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5. Unconditional Finiteness Theorems & Two-Sided CRT Sieves

While ceiling-saturated chains can be constructed up to $T(m) \approx W$ for $a_0 < 2^W$, once $T(m) > W$ (the **coasting regime**), we proved unconditional finiteness across the following regimes:

1. **Global Step-Rate & Even-Step Domination (`GlobalStepRateBounds.lean`, `ThreeAdicValuationConservation.lean`):** Every surviving orbit satisfies $s(k) \geq 2$, $s(k) = 2 \implies s(k+1) \geq 6, s(k+2) \geq 3, s(k+3) \geq 3$, and the global step-rate bound:

$$3m \leq T(m) + 1 \iff 2m \leq \text{evenStepSum } s\ m + 1.$$

   Moreover, the product of the first m block-boundary orbit values has exact 3-adic valuation $T(m) - m$:

$$\prod_{k=0}^{m-1} b_{k+1} = 3^{T(m)-m} \prod_{k=0}^{m-1} u(k), \qquad \gcd\left(\prod_{k=0}^{m-1} u(k),\ 6\right) = 1.$$

2. **Time-Independent Bounded-Block Escape Ceilings (`AperiodicBoundedBlockSqueeze.lean`, `TimeIndependentDeficiencySqueeze.lean`):** During any run of blocks bounded by $s(k+i) \leq c$ after $s(k-1) = 2$, the $2^{T(k)}$ factor cancels identically from `minSlackDeficiency`, bounding the run to at most **6 steps** for $c = 3$ and **4 blocks** for $c = 5$ regardless of how large $T(k)$ is!

3. **Repeated-Subword & Aperiodic Sturmian / Balanced Complexity Cutoffs (`RepeatedSubwordComplexity.lean`, `SturmianAndBalancedComplexityExclusion.lean`):** Whenever an L-block subword repeats at two origins $r_1 < r_2$ ($\forall i < L,\ s(r_1 + i) = s(r_2 + i)$), subtracting the multi-step quotient equations cancels the unknown remainder R_{win} identically and yields a common positive integer $K_0 \geq 1$ such that:

$$u(r_2) - u(r_1) = 2^{\text{tailSteps}^{(r_1)}(L-1)+1} K_0, \qquad u(r_2 + L - 1) - u(r_1 + L - 1) = 2 \cdot 3^{\text{blockSteps}^{(r_1)}(L-1)} K_0.$$

   Raising the resulting exponential bound $2^{T(r_2)+S+2} < 3^{T(r_2)+1}(2a_0 + 1)$ to the 5th ($3^5 = 243 < 2^8 = 256$) and 17th ($3^{17} < 2^{27}$) powers yields the **weighted logarithm-free cutoffs**:

$$5S \leq 3T(r_2) + 5W + 2, \qquad 17S \leq 10T(r_2) + 17W + 9, \qquad 51L \leq 10T(r_2) + 17W + 26.$$

   Combining these weighted inequalities with the **Finite Pigeonhole Principle on subword complexity** (`exists_repeated_subword_of_pattern_classifier`) unconditionally excludes substantial **aperiodic** classes—including ceiling-saturated classes ($s(k) \geq 3$) that never trigger any archimedean or deficiency pinch:

4. **Aperiodic Ceiling-Saturated $\{3,4\}$ & $[3,5]$ Sturmian Classes (`aperiodic_sturmian_three_four_exclusion`, `aperiodic_sturmian_three_five_exclusion`):** Any chain in $\{3,4\}$ (resp. $[3,5]$) of complexity $p(L) \leq L + C_0$ satisfies $3L \leq 12C_0 + 5W + 2$ (resp. $L \leq 50C_0 + 17W + 9$).

5. **General Sub-Critical Balanced Complexity (`balanced_subcritical_complexity_exclusion`):** For *any* average block length λ, any surviving chain with $10T(r_2) \leq (17 - \delta)S + D_0$ ($\delta \geq 1$, corresponding to complexity factor $\kappa \leq (17 - \delta)/10 < 1.7$) satisfies $3\delta L \leq D_0 + 17W + 9 + \delta$.

6. **Two-Sided Chinese Remainder Sieves & Quotient Floors (`Mod256CompleteSieve.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `Mod216TwoSidedQuotientFloor.lean`):**

7. **Mod-2304 (256×9) Two-Block Sieve:** Eliminates 2283 of 2304 residue classes of $(3^{s(k)}u(k) + 1) \bmod 2304$ (99.09%), locking the post-2 recovery branch to the single class $Y = 28 \pmod{2304}$.
8. **Unconditional Impossibility of $u(k) = 1$ ($k \geq 1$):** Looking one step backward ($k - 1 \to k$) mod 3 forces $u(k) = 1 \implies s(k)$ even, and looking one step forward ($k \to k + 1$) mod 4 forces $3^{s(k)} + 1 \equiv 2 \pmod 4 \implies s(k+1) = 1$ (immediate death).
9. **Mod-54, Mod-216 & Mod-1728 Quotient Floors:** Combining backward Mod-27 locks ($3 \leq s(k-1)$) with forward 2-adic locks forces $1735 \leq u(k)$ at $s(k) = 2$, $65 \leq u(k)$ across $2 \leq s(k) \leq 6$, and $103 \leq u(k)$ on ceiling-saturated runs ($3 \leq s(k) \leq 6$ with $3 \leq s(k+1)$).

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6. Certified Transcendence & Diophantine Barriers (Why the General Case Remains Open)

A central contribution of this formalization is proving **exact algebraic conservation laws (Dead Ends 1–18)** that explain why every standard transcendence and Diophantine method falls just short on a general aperiodic sequence s:

Barrier 1: Why Infinite Solutions Exist in $\mathbb{Z}_2 \times \mathbb{R}$ vs. Finite Positive Seeds (`TwoAdicCauchyConvergence.lean`, `MahlerSeriesHankelFramework.lean`, `FiniteSeedCarryAndBackwardDescent.lean`, `SubstitutionMahlerMethodAndGrandSynthesis.lean`)

The single power series $\Sigma_m(s) = \sum_{k=1}^{m} (2/3)^{T(k)+1} = \frac{2^{s(0)+1} R(s,m)}{3^{T(m)+1}}$ converges in **both** $\mathbb{R}$ (since $|2/3|_\infty = 2/3 < 1$) and $\mathbb{Q}_2$ (since $|2/3|_2 = 1/2 < 1$). Along any valid block chain (`dual_place_rational_limit_sign_split`):

$$0 < \Sigma_m(s) < \frac{1}{3}, \qquad -\frac{6a_0+2}{3} \le -\frac{2}{3} < 0, \qquad \Sigma_m(s) - \left(-\frac{6a_0+2}{3}\right) = \frac{2^{T(m+1)+1}u(m)}{3^{T(m)+1}}.$$

In $\mathbb{Z}_2 \times \mathbb{R}$, **infinite solutions genuinely exist** for every sequence s with minSlack $s\ m > 0$: there is a unique 2-adic integer $\alpha_2 \in \mathbb{Z}_2$ and real companion α_∞ satisfying all congruences and archimedean bounds. For a **positive integer seed** $a_0 \in \mathbb{N}$, the rational partial sums $\Sigma_m(s) \in (0, 1/3)$ converge in $\mathbb{Q}_2$ to the negative rational $-\frac{6a_0+2}{3} < 0$. *(Scope note: because $\mathbb{R}$ and $\mathbb{Q}_2$ are distinct completions of $\mathbb{Q}$, opposite real signs of the archimedean and 2-adic limits are not by themselves a contradiction for an arbitrary aperiodic series; they yield an immediate algebraic contradiction only when the series is forced to represent a common rational or algebraic relation in both places, as in the periodic macro-block case below.)* - **Periodic / Uniform Macro-Block Sign Obstruction (** `periodic_macro_block_seed_divisibility` **):** Whenever s has uniform macro-block step sum D and remainder R_0, the geometric series sums to the *same* positive rational $\frac{2^{s(0)+1}R_0}{3(3^D-2^D)} > 0$ in both $\mathbb{R}$ and $\mathbb{Q}_2$, forcing $2^{Dm} \mid \big((3^D-2^D)(3a_0+1) + 2^{s(0)}R_0\big)$. - **Concrete Resolution of the $G_{3,6}$ Product-Graph False Positive (** `g3_6_false_positive_cycle_and_real_orbit` **,** `constant_three_block_chain_eight_pow_dvd` **):** In `G_{3,6}`, the 3-cycle $(5,5) \to (4,2) \to (6,3) \to (5,5)$ ($s(k) = 3$) has valid fractional cycle $9/19 \to 4/19 \to 6/19 \to 9/19$ and 2-adic seed $a_0 = -9/19 \in \mathbb{Z}_2$, but any natural seed $a_0 \in \mathbb{N}$ must satisfy $8^m(19b_m + 9) = 27^m(19a_0 + 9)$, hence $8^m \mid (19a_0 + 9)$ and $3m \le W + 4$ for $a_0 < 2^W$. - **Backward-Lifting Descent Dichotomy (** `predecessor_lt_boundary` **,** `fracPart_backward_lift_identity` **,** `backward_archimedean_survival_dichotomy` **):** Every predecessor block boundary $a = (2^c u - 1)/3$ satisfies $a < b = 3^{c-1}u$ and $r_a = 1/3 + (2/3)^c r_b$, with $1/3 \le r_a < 13/27 < 1/2$ holding **unconditionally** for every $c \ge 3$.

Barrier 2: Rank-1 Determinant & Padé–Hankel Valuation Collapse (Dead Ends 12 & 18, `DeterminantFactorization.lean`, `MahlerSeriesHankelFramework.lean`)

When one forms the 2×2 determinant between steps r and $r+d$ to eliminate $2a_0 + 1$ (either from the archimedean slacks Δ_k or from the 2-adic numerators $P_k = 2^{T(k+1)+1}u(k)$), the determinant factors identically as:

$$3^{T(r+d)}\Delta_r - 3^{T(r)}\Delta_{r+d} = 3^{T(r)} \cdot 2^{T(r+1)+1} \cdot R(\text{shiftSeq } s\ r,\ d),$$

$$3^{T(r)+1}P_{r+d} - 3^{T(r+d)+1}P_r = 3^{T(r)+1} \cdot 2^{T(r+1)+1} \cdot R(\text{shiftSeq } s\ r,\ d).$$

Because $\gcd(R_d^{(r)}, 6) = 1$, the 2-adic valuation of the determinant drops from $T(r+d+1)+1$ all the way back to $T(r+1)+1$, preventing Padé approximation from gaining a 2-adic advantage over a single step.

Barrier 3: Baker–Matveev Linear Forms in Logarithms & Multi-Step GCDs (Dead Ends 13 & 17, `BakerMatveevFramework.lean`, `MultiStepQuotientGCD.lean`)

In `BakerMatveevFramework.lean`, we proved the exponentially small 3-logarithm linear form bounds:

$$0 < s(m+1)\log 2 - s(m)\log 3 + \log\left(\frac{u(m+1)}{u(m)}\right) < \frac{1}{3a_0+1}\left(\frac{2}{3}\right)^{T(m+1)},$$

$$0 < (T(m)+1)\log 3 - (T(m+1)+1)\log 2 + \log\left(\frac{2a_0+1}{u(m)}\right) \le \frac{1}{6a_0+2}\left(\frac{2}{3}\right)^{T(m)}.$$

When $u(m)$ is supported on a fixed finite set of primes (S-units), all algebraic bases are fixed, so invoking Matveev's external lower bound $\log|\Lambda| \ge -C(S, a_0)\log T(m)$ beats $-T(m)\log(3/2)$ and implies finiteness conditional on Matveev's theorem. However, in the general case: - Consecutive quotients are always coprime ($\gcd(u(m), u(m+1)) = 1$, `consecutive_u_coprime`). - Across d steps (`multi_step_u_gcd_eq_gcd_coprimeRemainder`), $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)}) < 3^{S_{r,d}}/2^{s(r)+1}$. - Even after canceling $\gcd(u(r), u(r+d))$, the reduced denominator satisfies $2^{T(r)}3^{S_{r,d}}\big(u(r)/\gcd(u(r), u(r+d))\big) > 3^{T(r)}(3a_0+1)$ (`multi_step_reduced_denominator_exponential_lower_bound`). Thus the logarithmic Weil height of the variable rational term grows linearly as $h \sim T(m)\log(3/2)$, causing Matveev's lower bound $-C_3 \cdot T(m)\log T(m)$ to lose to $-T(m)\log(3/2)$ by a factor of $\log T(m)$.

Barrier 4: Ridout's Theorem & the Critical Q_m^{-2} Exponent Identity (Dead End 14, `RidoutSubspaceFramework.lean`)

In the orbit equation $Q_m - P_m = \Delta_m$ with $Q_m = 3^{T(m)+1}(2a_0+1)$ and $P_m = 2^{T(m+1)+1}u(m)$, the $\{2,3\}$-free part of P_m is $u(m) \sim (3/2)^{T(m)}$ (with $2^{T(m)}$ in the denominator), while in any sub-ceiling phase $\Delta_m \le 2^{T(m)}$ (with $2^{T(m)}$ in the numerator). Multiplying $u(m)$ and Δ_m **cancels** $2^{T(m)}$ **identically** (`ridout_non_s_part_cancellation`):

$$2^{s(m)+D+1} \cdot u(m) \cdot \Delta_m < 3^{T(m)+1}(2a_0+1) = Q_m.$$

Consequently, the 3-place $\{\infty, 2, 3\}$ Ridout product satisfies (`ridout_three_place_quadratic_bound`):

$$\left(1 - \frac{P_m}{Q_m}\right) \cdot \frac{u(m)}{P_m} \cdot \frac{2a_0+1}{Q_m} < \frac{(6a_0+3)(2a_0+1)}{2^{s(m)+1}(6a_0+2)} \cdot \frac{1}{Q_m^2}.$$

Because the three place exponents sum to **exact** 2:

$$\underbrace{\left(1 - \frac{\log 2}{\log 3}\right)}_{|\cdot|_\infty} + \underbrace{\left(\frac{\log 2}{\log 3}\right)}_{|\cdot|_2} + \underbrace{1}_{|\cdot|_3} = 2,$$

the general bounded-block orbit sits **right on Ridout's critical Q_m^{-2} boundary**, crossing below $Q_m^{-2-\varepsilon}$ only when $s(m) \geq \delta T(m)$ or $\Delta_m \leq 2^{(1-\delta)T(m)}$.

Barrier 5: Multi-Origin Schlickewei Subspace Hyperplane Syzygy (Dead End 15, `MultiOriginSubspaceSyzygy.lean`)

Attempting to break the single-origin Q_m^{-2} tangency by combining $k \geq 3$ shifted origins $(0, r_1, r_2, \ldots)$ in Schlickewei's higher-dimensional p-adic Subspace Theorem fails because every scaled shifted slack $2^{T(r)}\Delta_{m-r}^{(r)}$ lies in the **same 2-dimensional plane** $\operatorname{span}\{\Delta_m^{(0)}, 3^{T(m)}\}$ as $m \to \infty$. Any three origins $(0, r_1, r_2)$ satisfy the exact m-independent hyperplane syzygy (`three_origin_subspace_syzygy`):

$$D_0(r_2) \cdot 3^{T(r_1)}\left(2^{T(r_1)}\Delta_{m-r_1}^{(r_1)} - \Delta_m^{(0)}\right) = D_0(r_1) \cdot 3^{T(r_2)}\left(2^{T(r_2)}\Delta_{m-r_2}^{(r_2)} - \Delta_m^{(0)}\right),$$

landing identically inside the degenerate exceptional subspace of Schlickewei's theorem.

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7. Quantitative Extensions of Dubickas (2008 / 2009): Shifted Carry Words, General Rational Bases p/q, and Explicit $O(\log \xi)$ Confinement Escape Bounds up to Total Length $25/36$

While Mahler's Z-number question concerns the specific half-interval $[0, 1/2)$ at base $3/2$, A. Dubickas (2008, *Math. Nachr.* **281**, 951–958; 2009, *Acta Arith.* **137**, 233–239) introduced a powerful relaxation based on the **shifted integer-part / carry recurrence**: for a coprime rational base $p/q > 1$ ($1 \leq q < p$, $\gcd(p, q) = 1$) and a real shift $\theta \in \mathbb{R}$, decomposing $\xi(p/q)^n + \theta = x_n + y_n$ with $x_n = \lfloor \xi(p/q)^n + \theta \rfloor \in \mathbb{Z}$ and $y_n = \{\xi(p/q)^n + \theta\} \in [0, 1)$ gives the exact dual identity (`dubBaseCarry_eq_fract_linear`):

$$s_n := qx_{n+1} - px_n = py_n - qy_{n+1} - (p - q)\theta \in \mathbb{Z}.$$

In `DubickasConfinementAndComplexityExtension.lean`, we formalized and substantially extended Dubickas's framework in five directions:

7.1 Multi-Step Carry Cancellation & the Sharp $\log q / \log(p/q)$ Sub-Critical Complexity Threshold

Iterating $qx_{n+1} = px_n + s_n$ over m steps yields the dual integer and fractional summation identities (`dubBaseX_multi_step_carry_identity`, `dubBaseY_multi_step_carry_identity`):

$$q^m x_{n+m} = p^m x_n + \sum_{t=0}^{m-1} p^{m-1-t} q^t s_{n+t}, \qquad q^m y_{n+m} = p^m y_n - \sum_{t=0}^{m-1} p^{m-1-t} q^t s_{n+t} - (p^m - q^m)\theta.$$

Whenever a length-L factor of s_n repeats at two indices $i < j$ ($\forall t < L,\ s_{i+t} = s_{j+t}$), subtracting the L-step identities cancels the carry sum identically (`dubBaseX_repeat_factor_cancellation`):

$$q^L(x_{j+L} - x_{i+L}) = p^L(x_j - x_i) \implies q^L \mid (x_j - x_i) \implies q^{L+j} < p^j q^W$$

for $\xi < q^W$ (`dubBaseX_repeated_window_exp_and_power_cutoff`). Raising $q^{L+j} < p^j q^W$ to the b-th power for any (a, b) satisfying $p^b < q^{a+b}$ yields the universal logarithm-free cutoff $bL + 1 \leq aj + bW$. Combining this with our general-base real logarithmic selection bridge (`exists_ab_of_lt_general_log_threshold`) proves (`no_subcritical_general_log_threshold_dubBaseCarry`): - **General-Base Sub-Critical Carry Complexity Theorem:** For **every** coprime pair $2 \leq q < p$ (including the open regime $p > q^2$), every $\xi \geq 2 + \frac{2q}{p-q}$, every $\theta \in [-1, 0]$, and **every** slope $\kappa < \frac{\log q}{\log(p/q)}$ (in particular, $\kappa < \frac{\log 2}{\log(3/2)} \approx 1.709511$ at base $3/2$ in `no_subcritical_log_threshold_dubCarry`), the shifted carry word cannot satisfy $\operatorname{prefixFactorCount}(s, L, C(L)) \leq C(L) \leq \kappa L + C_0$ at arbitrarily large lengths L. This strictly upgrades Dubickas's Sturmian carry-word contradiction ($\kappa = 1$) to the exact critical slope $\log q / \log(p/q)$.

7.2 Universal Two-Letter Alphabet, Finite Bispecial-Free Complexity $\leq L + 1$ & Linear-in-W Dubickas (2009) Escape

When $\{\xi(p/q)^n\} \in [s, s + 1/p]$ ($s \in [0, 1 - 1/p]$), setting $\theta = -s$ places $y_n = \{\xi(p/q)^n - s\} \in [0, 1/p]$, so $(p - q)s - q/p \leq s_n \leq (p - q)s + 1$. Since $0 < q/p < 1$, s_n is confined to the two-letter alphabet $\{\lfloor (p - q)s \rfloor,\ \lfloor (p - q)s \rfloor + 1\}$ (`dubBaseCarry_mem_two_letters`). Furthermore, if two occurrences of a length-$(m - 1)$ factor at i, j branched in both directions at their left and right endpoints, the m-step fractional identity would force $p^m + q^m \leq p^{m-1} + q^m$ ($p^m \leq p^{m-1}$, impossible for $p \geq 2$, `dubBaseCarry_no_bispecial_pair`). - **Finite Bispecial-Free Binary Complexity Theorem (`prefixFactorCount_le_add_one_of_no_bispecial`):** By partitioning the length-$(k + 1)$ factors in $[0, k + 1 + C)$ by their final bit and bounding the overlap of their length-k prefixes via the maximum differing index of any two distinct right-special prefixes, any binary word with no bispecial pair on $[0, L + C]$ satisfies $\operatorname{prefixFactorCount}(w, L, C) \leq L + 1$. - **Full General-Base Dubickas (2009) Theorem 1 & Strictly Linear $O(W)$ Escape (`dubickas2009_general_base_explicit_and_infinite_escape`, `dubickas2009_general_base_linear_in_W_escape`, `shifted_third_unconditional_explicit_escape`):** For **every** coprime $2 \leq q < p < q^2$ and **every** $s \in [0, 1 - 1/p]$, choosing (a, b) with $a < b$ and $p^b < q^{a+b}$ forces $\{\xi(p/q)^n\}$ to escape $[s, s + 1/p]$ within $n \leq 2bW + 2a + 2$ steps for all $\xi \in [2 + \frac{2q}{p-q}, q^W)$ (and within $n \leq W^2 + 3W + 2$ using $(a, b) = (1, 1)$ directly), and infinitely often for all $\xi > 0$. At base $3/2$ ($a = 3, b = 5, 3^5 < 2^8$), every $\xi \in [1, 2^W)$ escapes $[s, s + 1/3]$ by step $n \leq 5W + 14$. - **Floor-Residue Non-Stabilization & Open-Regime ($p > q^2$) Single-Cell Escape (`general_base_floor_mod_non_stabilization`, `general_base_non_straddling_cell_escape`):** Using $\lfloor x \rfloor \bmod m = r \iff \{x/m\} \in [r/m, (r + 1)/m)$, we prove that $\lfloor \xi(p/q)^n \rfloor \bmod m$ never stabilizes for any coprime $2 \leq q < p < q^2$ and $m \geq p$. Moreover, in the open regime $p > q^2$ ($p \geq 2q$, e.g. $5/2, 7/2, 7/3$), we prove that for *every* cell $[r/m, (r + 1)/m)$ that avoids the finitely many fixed points $\frac{k}{p-q}$ ($0 \leq k < p - q$), s_n is locked to a single constant k, forcing escape within $n \leq W + 4$ steps.

7.3 Human-Readable Inequality Proofs for Beyond-$1/p$ Single Windows at Bases $3/2$ and $4/3$

Dubickas (2008) showed that for certain shifts s, one can exceed the $1/p$ barrier on a single interval by forbidding short carry words: - **Base $3/2$ on $[1/6, 13/24)$ of Length $3/8 > 1/3$ (`shifted_three_eighths_carry_period_three` , `shifted_three_eighths_explicit_and_infinite_escape`):** Five linear fractional inequalities rule out `11` (2 steps), `000` (3 steps), `10101` (5 steps), `101001010` (9 steps), and `101001001` (9 steps), forcing $s_{k+3} = s_k$ (period 3) after step 2 and yielding an explicit $n \le W + 15$ escape bound. - **Base $4/3$ on $[1/3, 5/8)$ of Length $7/24 > 1/4 = 1/p$ (`base_four_three_carry_period_two` , `base_four_three_beyond_quarter_explicit_and_infinite_escape`):** Four linear fractional inequalities rule out `11` (2 steps: $16y_n - 9y_{n+2} = 7$), `000` (3 steps: $64y_n - 27y_{n+3} = 0$), `00100` (5 steps: $1024y_n - 243y_{n+5} = 144$), and `001010` (6 steps: $4096y_n - 729y_{n+6} = 891$), forcing $s_{k+2} = s_k$ (period 2) from step 2 onward and proving $Z_{4/3}(1/3, 5/8) = \emptyset$ with an explicit $n \le W + 8$ escape bound for $\xi \in [8, 3^W)$.

7.4 Verified Backward–Forward Integer Funnel Engine & Escape Bounds up to Total Length $25/36 > 2/3 > 1/2$

To certify explicit $O(W)$ escape bounds on multi-interval unions of total length well above $1/2$, we designed a **verified two-phase integer interval-arithmetic engine** (`inRatIntervalList_prev_of_funnelStepOk` , `inRatIntervalList_next_of_forwardStepOk`): 1. **Phase 1 (Backward Coarsened Funnel):** Starting from a scaled integer interval list T_0 at denominator D, `funnelStepOk D Tnext Tprev` checks that for every $(a, b) \in T_{\text{prev}}$, every carry $s \in \{-1, 0, 1, 2\}$, and every $(c, d) \in T_{\text{next}}$, the preimage $([a, b) \cap [(2c + sD)/3, (2d + sD)/3))$ is either empty or contained in a single interval of T_{prev}. Thus $y_0, \ldots, y_K \in T_0 \implies y_n \in T_K$ for all $n \le N - K$. 2. **Phase 2 (Forward Transient Pruning):** Because backward preimages retain transient intervals with in-degree 0 that eventually enter a cycle, `forwardStepOk D Tcur Tamb Tnext` pushes $y_n \in T_{\text{cur}}$ forward one step under $y_{n+1} = (3y_n - s)/2 \in T_{\text{amb}}$, eliminating all in-degree-0 transient components in 1–2 steps without any graph cycle search and collapsing the survivor set to a single fixed-carry or period-1 component!

Confinement Set / Family	Base p/q	Total Length	Carry Lock Mechanism	Explicit Finite Escape Bound	Lean Theorem
$[s,\ s + 1/p]$ ($s \in [0, 1 - 1/p]$)	$2 \le q < p < q^2$	$1/p$	No bispecial pair $\implies P(L) \le L + 1$	$n \le 2bW + 2a + 2$ ($W^2 + 3W + 2$)	`dubickas2009_general_base_linear_in_W_es`
$[s,\ s + 1/3]$ ($s \in [0, 2/3]$)	$3/2$	$1/3 \approx 0.3333$	No bispecial pair + $3^5 < 2^8$	$n \le 5W + 14$	`shifted_third_unconditional_explicit_esc`
$[r/m,\ (r+1)/m)$ non-straddling	$p \ge 2q \ge 4$	$1/m \le \frac{1}{2(p-q)}$	Single-letter carry lock $s_n = k$	$n \le W + 4$	`general_base_non_straddling_cell_escape`
$[1/3,\ 5/8)$	$4/3$	$7/24 \approx 0.2917 > 1/4$	4-inequality period-2 lock ($s_{k+2} = s_k$)	$n \le W + 8$	`base_four_three_beyond_quarter_explicit_a`
$[4/27,\ 1/2)$	$3/2$	$19/54 \approx 0.3519 > 1/3$	`11` , `101` , `000` ban ($s_{k+3} = s_k$)	$n \le W + 8$	`punctured_four_twenty_sevenths_period_thr`
$[8/57,\ 1/2)$	$3/2$	$41/114 \approx 0.3596 > 1/3$	Homoclinic catapult into $[4/19, 1/4)$	$n \le 3W + 20$	`punctured_eight_fifty_sevenths_explicit_a`
$[\frac{280}{2187}, \frac{648200}{4782969}) \cup [\frac{8}{57}, \frac{1}{2})$	$3/2$	$\frac{66729163}{181752822} \approx 0.3671$ (73.4% of $[0, \frac{1}{2})$)	7-step death band + $[8/57, 1/2)$ catapult	$n \le 3W + 27$	`two_island_punctured_half_explicit_and_in`
$[1/6,\ 13/24)$	$3/2$	$3/8 = 0.3750 > 1/3$	5-inequality period-3 lock ($s_{k+3} = s_k$)	$n \le W + 15$	`shifted_three_eighths_explicit_and_infini`
$[0,\ 1/5) \cup [4/5,\ 1)$	$3/2$	$2/5 = 0.4000$	1-step carry lock $s_{n+1} = s_n$	$n \le W + 5$	`nearest_int_fifth_explicit_escape`
$(8/39, 18/39) \cup (21/39, 31/39)$	$3/2$	$20/39 \approx 0.5128 > 1/2$	2-step `00` / `11` ban ($s_{k+2} = s_k$)	$n \le W + 8$	`dubickas2008_open_union_explicit_escape`
$[8/39, 18/39] \cup [21/39, 31/39]$	$3/2$	$20/39 \approx 0.5128 > 1/2$	Open interior + 4-cycle $2^m \mid (13x_r + v_r)$	$n \le 3W + 19$	`dubickas2008_closed_union_explicit_and_in`
4-interval union $U_{7/12}$	$3/2$	$7/12 \approx 0.5833 > 1/2$	7-step backward funnel ($D = 7296$)	$n \le W + 12$	`union_seven_twelfths_explicit_and_infinit`
5-interval union $U_{2/3}$	$3/2$	$2/3 \approx 0.6667 > 1/2$	9-step backward + 2-step forward ($D = 118098$)	$n \le W + 14$	`union_two_thirds_explicit_and_infinite_es`
6-interval union $U_{25/36}$	$3/2$	$25/36 \approx 0.6944 > 2/3$	11-step backward + 1-step forward ($D = 708588$)	$n \le W + 16$	`union_twenty_five_thirty_sixths_explicit_`
7-interval union $U_{17/24}$	$3/2$	$17/24 \approx 0.7083 > 70\%$	12-step self-refining + 1-step forward ($D = 472392$)	$n \le W + 17$	`union_seventeen_twenty_fourths_explicit_a`

7.5 The Thue–Morse Period-Doubling Cantor Cascade on $[c, 1 - c]$, Fermat-Denominator Reduction & the $[2/7, 5/7]$ Cutoff

For centered windows $[c, 1 - c]$ ($c \in (1/4, 2/5]$), the real hold set under $2y_{n+1} = 3y_n - s_n$ ($s_n \in \{0, 1\}$) is governed by a **Thue–Morse period-doubling Cantor cascade**: - **Fermat-Denominator Reduction & Mahler Product Identity (`tmCascadeCarryAcc_succ_factored` ,**

`tmCascadeCarryAcc_mahlerProd_identity`**):** Because the length-2^{k+1} Thue–Morse word decomposes as $T_{k+1} = T_k\overline{T_k}$, its carry accumulator $A(T_{k+1}) = \sum_{t=0}^{2^{k+1}-1} 3^{2^{k+1}-1-t}2^t\,\mathrm{tmBit}(t)$ factors as

$$A(T_{k+1}) = \left(3^{2^k} - 2^{2^k}\right)\left(A(T_k) + 2^{2^k}\right), \qquad \left(3^{2^k} - 2^{2^k}\right) - 2A(T_k) = \prod_{n=0}^{k-1}\left(3^{2^n} - 2^{2^n}\right).$$

Canceling $3^{2^{k-1}} - 2^{2^{k-1}}$ from $3^{2^k} - 2^{2^k}$ reduces the denominator of the period-2^k Thue–Morse cycle to the Fermat-like denominator $F_{k-1} := 3^{2^{k-1}} + 2^{2^{k-1}} \in \{5, 13, 97, 6817, 43112257, \ldots\}$. - **Closed-Form Entry Thresholds c_k, Difference-of-Squares Ratio & $[2/7, 5/7]$ Cutoff (** `tmCascadeEntry_closed_form`**,** `tmCascadeEntry_step_ratio_and_strictAnti`**,** `tmCascadeEntry_explicit_values_and_two_sevenths_cutoff`**,** `tmCascade_periodic_orbit_two_pow_dvd`**):** The minimum coordinate c_k of the period-2^k Thue–Morse cycle occurs at cyclic shift $2^{k-1}+1$ and satisfies the exact closed-form product identity and step ratio:

$$\left(3^{2^{k-1}} + 2^{2^{k-1}}\right)(4c_k - 1) = 3\prod_{n=0}^{k-2}\left(3^{2^n} - 2^{2^n}\right), \qquad \frac{4c_{k+1} - 1}{4c_k - 1} = \frac{3^{2^k} - 2^{2^k}}{3^{2^k} + 2^{2^k}} \in (0, 1).$$

Hence c_k strictly descends toward Dubickas's (2006) constant $c_\infty = \frac{1+\prod_{n\geq 0}(1-(2/3)^{2^n})}{4} \approx 0.28564732$, with first five rational values:

$$c_1 = \frac{2}{5} > c_2 = \frac{4}{13} > c_3 = \frac{28}{97} > c_4 = \frac{1948}{6817} > \frac{2}{7} > c_5 = \frac{12314908}{43112257},$$

where $c_4 - \frac{2}{7} = \frac{2}{47719} > 0$ and $\frac{2}{7} - c_5 = \frac{20158}{301785799} > 0$. Moreover, any real ξ landing on the level-k cycle at step r satisfies the 2-adic divisibility trap $2^{2^k m} \mid (F_{k-1}x_r + v)$ for all $m \geq 0$. - **Exact Match with Mahler's Binary Partition Sequence through Depth** 15 **(** `cantorTwoSevenths_matches_mahlerBinaryPartition_up_to_15_and_splits_at_16`**):** Because $c_5 < 2/7 < c_4$, the Cantor cylinder tree of $(2/7, 5/7)$ admits the first 4 levels of the Thue–Morse cascade ($L = 2, 4, 8, 16$) and matches Mahler's binary partition sequence $B(d) = B(d-1) + B(\lfloor d/2 \rfloor)$ identically for all $0 \leq d \leq 15$ (`[1, 2, 4, 6, 10, 14, 20, 26, 36, 46, 60, 74, 94, 114, 140, 166]`), before clipping the two mirror $k = 5$ (period-32) branches at depth $d = 2^4 = 16$ (`200` vs. $B(16) = 202$).

7.6 The General Gap-g Window-to-Residue Ladder ($g \geq 3$) & General g-Rung First-Defect Valuation Lock

For bottom-anchored windows $[0, a)$ ($g \geq 3$), defining $\alpha_g = \frac{3^{g-1}}{3^g - 2^g}$ and $\beta_g = \frac{1+(2/3)^{g-1}}{3} = \frac{3^{g-1}+2^{g-1}}{3^g}$, we formalized the general g-rung ladder and its 2-adic valuation lock: - **Strict Rung Ordering & Short-Gap Fractional Cost (** `rungAlpha_lt_rungBeta_lt_half`**,** `gap_ladder_short_gap_cost_and_cap`**):** Since $2^g < 3^{g-1}$ for all $g \geq 3$, every rung satisfies $\frac{1}{3} < \alpha_g < \beta_g < \frac{1}{2}$ with $\beta_g - \alpha_g = \frac{2^{g-1}(3^{g-1}-2^g)}{3^g(3^g-2^g)} > 0$. Any odd step followed by another odd step at distance $1 \leq d \leq g-1$ has fractional part $\geq \frac{1}{3} + (2/3)^d \cdot \frac{1}{3} \geq \beta_g$, whereas any block of length $c \geq g$ maps $[0, \alpha_g]$ into $[0, \alpha_g]$. - **Universal Rung-g Affine Invariant $I_g(b) = (3^g - 2^g)b + 3^{g-1}$ & First-Defect Valuation Lock (** `rungAffInv_single_block_identity`**,** `rungAffInv_pure_prefix_identity`**,** `general_rung_first_defect_valuation_lock`**):** Across any block $3b+1 = 2^c u$, $I_g(b)$ satisfies the universal identity $3I_g(b) = (3^g - 2^g)2^c u + 2^g$. Consequently, m consecutive blocks of length g force $2^{gm} I_g(b_m) = 3^{gm} I_g(a_0)$, and a first defect $s(j_0) \geq g+1$ after j_0 pure blocks of length g permanently locks the 2-adic valuation of $(3^g - 2^g)a_0 + 3^{g-1}$ to **exactly** $g(j_0+1)$:

$$2^{g(j_0+1)} \,\Big|\, \left((3^g - 2^g)a_0 + 3^{g-1}\right) \quad \text{and} \quad 2^{g(j_0+1)+1} \nmid \left((3^g - 2^g)a_0 + 3^{g-1}\right).$$

7.7 The Punctured $[0, 1/2)$ Homoclinic Catapult Trap on $[8/57, 1/2)$ & the $\varepsilon^* = 8/57$ Phase Transition

When Mahler's interval $[0, 1/2)$ is punctured at the origin to $[\varepsilon, 1/2)$, the unbounded `0^g` staircase near 0 is cut off and the Cantor cylinder tree undergoes an exact homoclinic phase transition at $\varepsilon^* = \frac{8}{57} = \frac{2}{3} \cdot \frac{4}{19} \approx 0.140351$: - **Pure Period-3 Lock on** $[4/27, 1/2)$ **(** `punctured_half_step_dichotomy_and_forbidden_short_gaps`**,** `punctured_four_twenty_sevenths_period_three_lock_and_escape`**):** For any $\varepsilon \geq 1/16$, `11` and `101` are forbidden on $[\varepsilon, 1/2)$. When $\varepsilon \geq 4/27 \approx 0.14815$, three consecutive zeros (`000`) are also forbidden (`8 y_{k+3} = 27 y_k \ge 4`), forcing $s_k + s_{k+1} + s_{k+2} = 1$ and $s_{k+3} = s_k$ for all $k \geq 0$, with explicit escape in $n \leq W + 8$ steps. - **The Homoclinic Catapult Trap on** $[8/57, 1/2)$ **(** `punctured_eight_fifty_sevenths_catapult_and_absorbing_trap`**,** `punctured_eight_fifty_sevenths_explicit_and_infinite_escape`**):** On $[8/57, 1/2)$ (length $41/114 \approx 0.3596 > 1/3$), four consecutive zeros (`0000`) are forbidden, and any three-zero block (`000`) at step k forces $s_{k+3} = 1$ and **catapults** y_{k+4} into the absorbing period-3 trap $[4/19, 1/4)$:

$$16y_{k+4} = 81y_k - 8 \geq 81 \cdot \frac{8}{57} - 8 = \frac{64}{19} \implies \frac{4}{19} \leq y_{k+4} < \frac{1}{4}.$$

Once $y_m \in [4/19, 1/4)$, every subsequent 3-step block is forced to be `(001)` with $y_{m+3} \in [4/19, 1/4)$, locking $s_{t+3} = s_t$ for all $t \geq m-1$. Thus at most **one** `000` block can occur in an entire orbit, and every $\xi \in [1, 2^W)$ escapes $[8/57, 1/2)$ in at most $n \leq 3W + 20$ steps ($Z_{3/2}([8/57, 1/2)) = \emptyset$). - **Sub-**$8/57$ **Homoclinic Periodic Cascade & Kernel-Certified Cylinder Counts (** `homoclinicCycleMin_closed_form_and_strictMono`**,** `cantorPuncturedHalf_phase_transition_counts`**):** Below $\varepsilon^* = 8/57$, the periodic cycles $\mathcal{O}_m = \texttt{(001)}^m\texttt{(0001)}$ of period $3m+4$ appear at minimum coordinates

$$\varepsilon_m = \frac{8}{57} - \frac{4 \cdot 2^{3m+4}}{57\left(3^{3m+4} - 2^{3m+4}\right)} \quad \left(\varepsilon_1 = \frac{280}{2059} < \varepsilon_2 = \frac{8072}{58025} < \varepsilon_3 = \frac{222040}{1586131} < \cdots \uparrow \frac{8}{57}\right).$$

Kernel evaluation (`by decide`) certifies the Cantor cylinder counts $|C_d([\varepsilon, 1/2))|$ for $0 \leq d \leq 10$ across all four regimes: constant 3 at $\varepsilon = 4/27$ (`[1, 2, 3, 3, ..., 3]`), constant 6 at $\varepsilon = 1/7$ (`[1, 2, 3, 4, 5, 6, 6, ..., 6]`), exact $d+1$ linear growth at $\varepsilon^* = 8/57$ (`[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`), and bifurcation above $d+1$ at depth $d = 7$ (period 7 of $\mathcal{O}_1$) for $\varepsilon = 1/8 < \varepsilon_1$ (`[1, 2, 3, 4, 5, 6, 7, 9, 11, 13, 16]`).

7.8 The Universal Rung-g Catapult Staircase ($\varepsilon_g^* \to 0$) and the Leak-Free Inductive Elevator

Section 7.7's catapult trap at $g=3$ ($\varepsilon_3^* = 8/57$) extends to an infinite inductive staircase across **all rungs** $g \geq 3$ whose left boundary ε_g^* converges to 0: - **Universal Four-Term Rung Ordering, Corridor Overlap & Geometric Decay (`rungCatapult_ordering_and_corridor_overlap`):** For every $g \geq 3$, define the period-g post-reset fixed point $z_g^* = \frac{2^{g-1}}{3^g - 2^g}$, the $(g+1)$-defect boundary $b_g = \frac{2^{g-1}}{3^g}$, and the Rung-$g$ catapult floor $\varepsilon_g^* = \frac{2^g}{3(3^g-2^g)} = \frac{2}{3}z_g^*$. Because $2^g < 3^{g-1}$ for all $g \geq 3$, these satisfy the strict inequality chain

$$0 < \varepsilon_{g+1}^* < z_{g+1}^* < \varepsilon_g^* < b_g < z_g^* < b_{g-1}, \qquad \varepsilon_g^* < \left(\frac{2}{3}\right)^g \xrightarrow{g\to\infty} 0,$$

with explicit rational triples $(\varepsilon_3^*, b_3, z_3^*) = (8/57, 4/27, 4/19)$, $(\varepsilon_4^*, b_4, z_4^*) = (16/195, 8/81, 8/65)$, and $(\varepsilon_5^*, b_5, z_5^*) = (32/633, 16/243, 16/211)$. Because $\varepsilon_g^* < b_g$, adjacent Rung-g corridors $J_g = [\varepsilon_g^*, b_{g-1})$ overlap on $[\varepsilon_g^*, b_g)$ and tile $(0, 2/9)$ all the way down to 0. - **Universal Rung-g Catapult & Absorbing Upward Expansion (`universal_rung_catapult_and_upward_expansion`):** Under the k-step post-reset return map $T_k(z) = (3/2)^k z - 1/2$, every rung $g \geq 3$ satisfies

$$T_g(z) - z_g^* = \left(\frac{3}{2}\right)^g (z - z_g^*), \qquad T_{g+1}(z) - z_g^* = \left(\frac{3}{2}\right)^{g+1} (z - \varepsilon_g^*).$$

Hence any $(g+1)$-step block from $z \geq \varepsilon_g^*$ **catapults** $T_{g+1}(z) \geq z_g^* > b_g$, and any g-step block from $z \geq z_g^*$ expands upward ($T_g(z) \geq z \geq z_g^*$). - **The Leak-Free Inductive Elevator Across Rungs** $3 \leq g \leq G$ **(`leak_free_inductive_elevator_monotone_rung_descent`):** Each interval $[b_g, b_{g-1})$ splits into the **Rung-g leak window** $L_g = [b_g, z_g^*)$ of width $\mathrm{rungLeakWidth}(g) = \frac{2^{2g-1}}{3^g(3^g-2^g)} \leq \frac{32}{513}\left(\frac{4}{9}\right)^{g-3}$ and the **Rung-g upward trap** $\mathrm{Trap}_g = [z_g^*, b_{g-1})$. Whenever a post-reset trajectory evolves in the leak-free union $\mathcal{U}_G = [\varepsilon_G^*, b_G) \cup \bigcup_{g=3}^{G} [z_g^*, b_{g-1})$ (`ValidLeakFreeElevator G z s`), at most one initial catapult step ($s_0 = G+1$) occurs at $m=0$, after which $z_1 \leq z_2 \leq \cdots$ is monotonically non-decreasing and the block lengths satisfy $G+1 \geq s_0 \geq s_1 \geq s_2 \geq \cdots \geq 3$—forcing one-way upward promotion toward Rung 3 regardless of how close ε_G^* is to 0.

7.9 Self-Cleaning Death-Band Decomposition of Leak Windows $L_g = [b_g, z_g^*)$ & the $280/2059$ Seven-Step Catapult-Death Trap

Even without manually puncturing $L_g = [b_g, z_g^*)$, almost the entire leak window is **automatically self-cleaning** on $[\varepsilon, 1/2)$: - **The m-Step Automatic Death-Band Ladder $D_{g,m}(\varepsilon) \subset [b_g, z_g^*)$ (`rungDeathBand_step_and_measure_decomposition`):** For every rung $g \geq 3$ and depth $m \geq 1$, the m-step preimage of $[0, \varepsilon)$ under $T_g(w) = (3/2)^g w - 1/2$ is the interval $D_{g,m}(\varepsilon) = [\mathrm{rungDeathBandLo}(g,m),\ \mathrm{rungDeathBandHi}(g,m,\varepsilon))$ with

$$\mathrm{rungDeathBandLo}(g,m) = z_g^*\left(1 - \left(\frac{2}{3}\right)^{gm}\right), \qquad \mathrm{rungDeathBandHi}(g,m,\varepsilon) = \mathrm{rungDeathBandLo}(g,m) + \left(\frac{2}{3}\right)^{gm}\varepsilon.$$

At $m=1$, $\mathrm{rungDeathBandLo}(g,1) = b_g$, and every $w \in D_{g,1}(\varepsilon)$ falls into $[0,\varepsilon)$ in a single g-step block ($0 \leq T_g(w) < \varepsilon$). For all $m \geq 1$, T_g maps $D_{g,m+1}(\varepsilon)$ onto $D_{g,m}(\varepsilon)$, and for $0 \leq \varepsilon \leq b_g$ these bands are pairwise disjoint inside $[b_g, z_g^*)$ with total geometric-series measure $\mathrm{rungCleanedMeasure}(g,\varepsilon) = \frac{2^g}{3^g - 2^g}\varepsilon$ and residual uncleaned measure

$$\mathrm{rungResidualLeakMeasure}(g,\varepsilon) = \mathrm{rungLeakWidth}(g) - \frac{2^g}{3^g-2^g}\varepsilon = \frac{2^g}{3^g-2^g}(b_g - \varepsilon).$$

At $\varepsilon = \varepsilon_3^* = 8/57$ ($g=3$), the automatic death bands clean out $\frac{64}{1083}$ out of $|L_3| = \frac{32}{513}$—an exact fraction $\frac{18}{19} \approx 94.74\%$ of the Rung-3 leak window—leaving a residual uncleaned measure of only $\frac{32}{9747} \approx 0.003283$. - **The 7-Step Catapult-into-Death-Band Trap Below $8/57$ (`catapult_into_death_band_seven_step_escape`):** For $\varepsilon \geq 1/8$, any point $y_k \in \left[\frac{280}{2187},\ \frac{280+128\varepsilon}{2187}\right)$ is forced to take the 4-step block `0001` into the 1-step death band $D_{3,1}(\varepsilon) = \left[\frac{4}{27},\ \frac{4+8\varepsilon}{27}\right)$, followed by the 3-step block `001` into $[0,\varepsilon)$—dying in $4+3=7$ steps (`0001001`)! Moreover, $\varepsilon < \frac{280+128\varepsilon}{2187} \iff \varepsilon < \frac{280}{2059} = \varepsilon_1$, explaining mechanically why $\varepsilon_1 = \frac{280}{2059}$ is the first homoclinic threshold below $\frac{8}{57}$.

7.10 Higher-Order Gap-to-Trigger Bijection ($\mathrm{Gap}_{g,m}(\varepsilon) \xrightarrow{T_g^m} [\varepsilon, b_g)$), K-Pass Decay to 0, and the $2^{7m} \mid (2059x_r + 280)$ Leak-Threading Survivor Lock

What happens to the remaining uncleaned gaps $\mathrm{Gap}_{g,m}(\varepsilon) := [\mathrm{rungDeathBandHi}(g,m,\varepsilon),\ \mathrm{rungDeathBandLo}(g,m+1))$ between consecutive death bands $D_{g,m}(\varepsilon)$ and $D_{g,m+1}(\varepsilon)$? - **Universal Gap-to-Trigger Bijection & K-Pass Measure Decay (`rungDeathGap_maps_onto_catapult_zone`):** Under the m-fold return map $T_g^{[m]} = \mathrm{rungReturnMapIter}(g,m,\cdot)$, every uncleaned gap $\mathrm{Gap}_{g,m}(\varepsilon)$ ($g \geq 3, m \geq 1$) satisfies the displacement law $T_g^{[m]}(w) - z_g^* = (3/2)^{gm}(w - z_g^*)$, has exact width $(2/3)^{gm}(b_g - \varepsilon)$, and maps **bijectively onto the catapult trigger zone** $[\varepsilon, b_g)$:

$$T_g^{[m]}\big(\mathrm{rungDeathBandHi}(g,m,\varepsilon)\big) = \varepsilon, \qquad T_g^{[m]}\big(\mathrm{rungDeathBandLo}(g,m+1)\big) = b_g,$$

$$w \in \mathrm{Gap}_{g,m}(\varepsilon) \iff T_g^{[m]}(w) \in [\varepsilon, b_g).$$

Consequently, every point surviving the first pass through L_g is driven back into $[\varepsilon, b_g)$ and stretched by T_{g+1} across the death bands $D_{g,k}(\varepsilon)$ a second, third, ..., K-th time (at $g=3, \varepsilon = 8/57$, the single-pass survival ratio $\frac{1}{19}$ decays as $(1/19)^2 = \frac{1}{361}$ and $(1/19)^3 = \frac{1}{6859} \to 0$, eliminating 100% of the width in the $K \to \infty$ limit). - **Real 7-Cycle Inside $\mathrm{Gap}_{3,1}(\varepsilon_1)$, $2^{7m} \mid (2059x_r + 280)$ Lock & Constructive Leak-Threading Survivors (`homoclinic_seven_cycle_in_gap_and_divisibility_lock`):** At $\varepsilon_1 = \frac{280}{2059}$, the post-`0001` point $w = \frac{388}{2059} = T_4(\varepsilon_1) = \mathrm{rungDeathBandHi}(3,1,\varepsilon_1) \in \mathrm{Gap}_{3,1}(\varepsilon_1) = \left[\frac{388}{2059},\ \frac{140}{729}\right)$ returns to $T_3(w) = \frac{280}{2059} = \varepsilon_1$, generating the real 7-cycle $\mathcal{O}_1 =$ `(0001)(001)` inside $[\frac{280}{2059}, \frac{1}{2})$. However, any Mahler trajectory $\xi(3/2)^n = x_n + y_n$ with $y_r = y_{r+7m} = \frac{280}{2059}$ satisfies the 2-adic divisibility lock

$$\left(2^{7m} : \mathbb{Z}\right) \;\Big|\; (2059x_r + 280),$$

ruling out $m = \infty$ for all positive seeds while constructively producing explicit finite-horizon leak-threading survivors: $x_0 = 56 \pmod{2^7}$, $x_0 = 1464 \pmod{2^{14}}$, and $x_0 = 1443256 \pmod{2^{21}}$.

7.11 Universal Homoclinic Lock Family $2^{(3k+4)m} \mid (D_k x_r + N_k)$ & Mutually Incompatible Multi-Defect Valuation Locks

We generalize our 2-adic locks across the entire infinite homoclinic cascade and across consecutive multi-defect transitions: - **Universal Homoclinic Cycle Family Lock (`homoclinic_family_cycle_divisibility_lock`):** For every $k \in \mathbb{N}$ in the homoclinic cascade $\mathcal{O}_k = \texttt{(001)}^k\texttt{(0001)}$ approaching $\frac{8}{57}$ from below, the integer numerator $N_k = \text{homoclinicCycleNum}(k)$ and denominator $D_k = \text{homoclinicCycleDen}(k) = 3^{3k+4} - 2^{3k+4}$ satisfy

$$19N_k = 8\left(27^{k+1} - 8^{k+1}\right), \qquad \varepsilon_k = \text{homoclinicCycleMin}(k) = \frac{N_k}{D_k},$$

with $(N_1, D_1) = (280, 2059)$, $(N_2, D_2) = (8072, 58025)$, and $(N_3, D_3) = (222040, 1586131)$. Whenever $\text{dubY}\ \xi\ 0\ r = \text{dubY}\ \xi\ 0\ (r + (3k+4)m) = \varepsilon_k$, the integer part $x_r = \text{dubX}\ \xi\ 0\ r$ satisfies:

$$\left(2^{(3k+4)m} : \mathbb{Z}\right) \;\Big|\; (D_k x_r + N_k).$$

- **Universal Rung-g Defect Cocycle & Incompatible Multi-Defect Valuation Locks (`rungAffInv_general_defect_step_identity`, `exactTwoPowLock_incompatible_of_ne`, `general_rung_two_defect_simultaneous_lock`):** Across any defect block $3b + 1 = 2^c u$ ($c \geq g \geq 1$) with $b' = \text{orbit}\ b\ c = 3^{c-1}u$, the Rung-$g$ affine invariant $I_g(b) = (3^g - 2^g)b + 3^{g-1}$ satisfies:

$$2^c I_g(b') + 2^g 3^{g-1}\left(3^{c-g} - 2^{c-g}\right) = 3^c I_g(b).$$

Consequently, if a trajectory has j_0 pure blocks of length g, a first defect $c = s(j_0) \geq g+1$, j_1 pure blocks of length g, and a second defect $s(j_0 + 1 + j_1) \geq g + 1$, then a_0 must **simultaneously** satisfy: 1. $\text{ExactTwoPowLock}\big(g(j_0+1),\ I_g(a_0)\big)$, and 2. $\text{ExactTwoPowLock}\Big(gj_0 + c + g(j_1+1),\ 3^{gj_0+c} I_g(a_0) - 2^{g(j_0+1)} 3^{g-1}\left(3^{c-g} - 2^{c-g}\right)\Big)$, where $\text{ExactTwoPowLock}(e, I) :\iff 2^e \mid I \wedge 2^{e+1} \nmid I$. Because $\text{ExactTwoPowLock}(e_1, I)$ and $\text{ExactTwoPowLock}(e_2, I)$ are **mutually incompatible** whenever $e_1 \neq e_2$, no starting seed a_0 can be compatible with two distinct first-defect horizons $j_0 \neq j_0'$ OR two distinct second-defect horizons $j_1 \neq j_1'$.

7.12 The Two-Island Punctured $[0, 1/2)$ Theorem Reaching $\frac{280}{2187}$ (73.43% of $[0, 1/2)$), the $17/24 > 70\%$ Seven-Interval Union, and the Arbitrary K-Defect Valuation Tower

- **Two-Island Punctured $[0, 1/2)$ Theorem Reaching Down to $\frac{280}{2187} \approx 0.12803$ (`two_island_punctured_half_explicit_and_infinite_escape`):** Combining Section 7.7's catapult trap on $[8/57, 1/2)$ with Section 7.9's 7-step catapult-into-death-band trap at $\varepsilon = \frac{280}{2187} \approx 0.12803$ ($\frac{280+128\varepsilon}{2187} = \frac{648200}{4782969} \approx 0.13552$), we prove that on the two-island subset of $[0, 1/2)$:

$$\mathcal{W}_{280/2187} = \left[\frac{280}{2187},\ \frac{648200}{4782969}\right) \;\cup\; \left[\frac{8}{57},\ \frac{1}{2}\right)$$

  (which removes only a gap of width $\frac{438784}{90876411} \approx 0.004828 < \frac{1}{200}$ below $\frac{8}{57}$ and covers $\frac{66729163}{181752822} \approx 0.367142$, or 73.43% **of** $[0, 1/2)$), every $\xi \in [1, 2^W)$ escapes in at most $n \leq 3W + 27$ **steps** ($Z_{3/2}(\mathcal{W}_{280/2187}) = \emptyset$).
- **Self-Refining Funnel Engine & the $17/24 = 51/72 \approx 0.70833 > 70\%$ Seven-Interval Union (`inRatIntervalList_terminal_of_selfFunnelChainOk`, `union_seventeen_twenty_fourths_explicit_and_infinite_escape`):** By refining both endpoints against the current funnel level T_k (`selfFunnelChainOk` checking `funnelStepOk D Tk Tk Tk1`), we certify in 12 backward steps and 1 forward step over $D = 472392$ (at most 8 intervals per level) that the 7-interval union

$$U_{17/24} = \left[0, \frac{1}{12}\right) \cup \left[\frac{1}{9}, \frac{11}{36}\right) \cup \left[\frac{4}{9}, \frac{2}{3}\right) \cup \left[\frac{25}{36}, \frac{3}{4}\right) \cup \left[\frac{7}{9}, \frac{19}{24}\right) \cup \left[\frac{5}{6}, \frac{8}{9}\right) \cup \left[\frac{11}{12}, 1\right)$$

  of total length $17/24 = 51/72 \approx 0.708333 > 70\%$ forces every $\xi \in [1, 2^W)$ to escape in at most $n \leq W + 17$ **steps**.
- **Arbitrary K-Defect Simultaneous Valuation Lock Tower (`rungDefectCocycle_tower_identity`, `general_rung_multi_defect_valuation_tower`):** Along any `ValidBlockChain a_0 s u K` with $s(k) \geq g \geq 1$ for all $k < K$, defining $\text{rungDefectCocycle}(g, s, m)$ recursively via $\text{rungDefectCocycle}(g, s, m+1) = 3^{s(m)}\text{rungDefectCocycle}(g, s, m) + 2^{\text{blockSteps}(s,m)+g} 3^{g-1}(3^{s(m)-g} - 2^{s(m)-g})$, every horizon $m \leq K$ satisfies

$$2^{\text{blockSteps}(s,m)} \cdot I_g(b_m) + \text{rungDefectCocycle}(g, s, m) = 3^{\text{blockSteps}(s,m)} \cdot I_g(a_0),$$

  and at **every** defect index $m < K$ with $s(m) \geq g+1$ simultaneously, the seed a_0 satisfies $\text{ExactTwoPowLock}\big(\text{blockSteps}(s,m) + g,\ 3^{\text{blockSteps}(s,m)} I_g(a_0) - \text{rungDefectCocycle}(g, s, m)\big)$.

7.13 Pushing Into the Cantor Dust Below $8/57$: The k-Block Minimum Defect Spacing Law, Forced `0001001` Macro-Blocks on $[280/2059, 1/2)$, and the $(3, 3k+4)$ Perron–Fibonacci Hierarchy

What is the exact combinatorial and dynamical structure of the zero-width Cantor dust when ε drops below $\varepsilon^* = \frac{8}{57}$ through the homoclinic cascade $\varepsilon_k = \frac{N_k}{D_k}$ ($k \geq 1$)? - **Universal k-Block Minimum Defect Spacing Law (`cantor_dust_minimum_defect_spacing_law`):** For every $k \geq 1$ and every post-

reset point $z \geq \varepsilon_k = \text{homoclinicCycleMin}(k)$, after a 4-step defect block `0001` ($z_1 = T_4(z)$), every intermediate 3-step return $0 \leq r < k$ stays strictly above the Rung-3 defect boundary $b_3 = \frac{4}{27}$:

$$\text{rungReturnMapIter}\big(3, r, \text{rungReturnMap}(4, z)\big) \geq \frac{388}{2059} > \frac{4}{27} = \text{rungBoundary}(3),$$

and the k-th return satisfies $\text{rungReturnMapIter}\big(3, k, \text{rungReturnMap}(4, z)\big) \geq \varepsilon_k$. Consequently, on $[\varepsilon_k, 1/2)$, **every `0001` defect block is forced to be followed by at least k pure `001` blocks** before another `0001` block can occur—forming a composite macro-block `0001(001)`^{k} of length $3k+4$. - **Direct 7-Step Macro-Block Lock on $[280/2059, 1/2)$ (`punctured_eps1_forced_seven_step_macro_block`):** On $[\varepsilon_1, 1/2) = [\frac{280}{2059}, \frac{1}{2})$, any `000` at step n forces the full 7-step carry word $(s_n, \ldots, s_{n+6}) = (0,0,0,1,0,0,1)$ (`0001001`), with $\frac{388}{2059} \leq y_{n+4} < \frac{1}{4}$ and $\frac{280}{2059} \leq y_{n+7} < \frac{1}{4}$—forbidding two consecutive 4-blocks (`0001000`). - **Kernel-Certified $(3, 3k+4)$ Perron–Fibonacci Cantor Cylinder Hierarchy & Characteristic Polynomial Root Squeeze (`cantorDust_perron_fibonacci_counts`, `perronCharPoly_sign_change_and_root_squeeze`):** Because the Cantor dust on $(\varepsilon_{k-1}, \varepsilon_k]$ is generated by macro-blocks of lengths 3 (`001`) and $3k+4$ (`0001(001)`^{k}), its cylinder first-differences $\Delta C_k(d) = |C_{d+1}| - |C_d|$ satisfy the $(3, 3k+4)$ Perron–Fibonacci recurrence $\Delta C_k(d) = \Delta C_k(d-3) + \Delta C_k(d-(3k+4))$, kernel-certified (`by decide`) through depth $d = 14$ at $k = 1$ ($L_1 = 7$, `[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 19, 22]`) and at $k = 2$ ($L_2 = 10$, `[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 17]`). Moreover, its characteristic polynomial $P_k(\lambda) = \lambda^{3k+4} - \lambda^{3k+1} - 1 = \lambda^{3k+1}(\lambda^3 - 1) - 1$ satisfies $P_k(1) = -1 < 0$ and $P_k(1 + 1/m) > 0$ whenever $m^2 \leq 3(3k+1)$, squeezing the Perron growth rate λ_k into $(1, 1 + 1/m)$ and proving $\lambda_k \to 1^+$ (topological entropy $\log \lambda_k \to 0$) as $k \to \infty$. - **Universal Catapult-Death Band Ladder $C_m(\varepsilon) = \left[\frac{N_m}{3^{3m+4}}, \frac{N_m + 2^{3m+4}\varepsilon}{3^{3m+4}}\right)$ & Homoclinic Tangency (`universal_catapult_death_band_ladder`):** For every $m \geq 1$, the `0001` preimage $C_m(\varepsilon) := T_4^{-1}(D_{3,m}(\varepsilon))$ of the m-th Rung-3 death band has lower endpoint $\text{catapultDeathLo}(m) = \frac{N_m}{3^{3m+4}}$ and upper endpoint $\text{catapultDeathHi}(m, \varepsilon) = \frac{N_m + 2^{3m+4}\varepsilon}{3^{3m+4}}$, sharing the exact numerator $N_m = \text{homoclinicCycleNum}(m)$ of $\varepsilon_m = \frac{N_m}{3^{3m+4} - 2^{3m+4}}$. Every point in $C_m(\varepsilon)$ falls into $[0, \varepsilon)$ in $3m+4$ steps (`0001 (001)^m`), $\text{catapultDeathHi}(m, \varepsilon_m) = \varepsilon_m$ (with $\varepsilon < \text{catapultDeathHi}(m, \varepsilon) \iff \varepsilon < \varepsilon_m$), and for $0 \leq \varepsilon \leq \frac{4}{27}$ the bands $C_m(\varepsilon)$ are pairwise disjoint inside $[\frac{280}{2187}, \frac{8}{57})$ with total measure $\frac{128}{1539}\varepsilon$—bringing the combined measure of $[8/57, 1/2) \cup \bigcup_{m=1}^{\infty} C_m(\frac{280}{2187})$ to $\frac{2492689}{6731586} \approx 0.370297$, or 74.06% **of** $[0, 1/2)$. - **Hidden Arithmetic Structure in the Cantor Dust: $\{3,4\}$ Block Confinement on $[8/81, 1/2)$, Defect-to-Defect Recurrence $2^{3j_1+4}U_1 + 9 = 3^{3j_0+4}U_0$, Mod-8 Lock & Sparse-Defect Descent (`punctured_eight_eighty_firsts_forbids_four_zeros_and_short_gaps`, `cantor_dust_defect_to_defect_quotient_recurrence`, `cantor_dust_mod_eight_lock_and_sparse_descent`, `no_infinite_sparse_defect_cantor_dust_chain`):** On $[\frac{8}{81}, \frac{1}{2}) \approx [0.098765, 0.5)$ (80.25% of $[0, 1/2)$), `11`, `101`, and `0000` are forbidden, so every block has length $s(m) \in \{3, 4\}$. Across consecutive `0001` defects separated by j_0 and j_1 blocks of 3, the odd defect quotients U_0, U_1 ($19a_0 + 9 = 2^{3j_0+3}U_0$, $19b_{j_0+1} + 9 = 2^{3j_1+3}U_1$) satisfy:

$$2^{3j_1+4} \cdot U_1 + 9 = 3^{3j_0+4} \cdot U_0.$$

This forces $16 \mid (3^{3j_0+4}U_0 - 9)$ and $(3^{j_0}U_0) \bmod 8 = 1$—excluding $U_0 \bmod 8 \in \{5, 7\}$ and locking ($j_0 \bmod 2 = 0 \leftrightarrow U_0 \bmod 8 = 1$) and ($j_0 \bmod 2 = 1 \leftrightarrow U_0 \bmod 8 = 3$). Moreover, whenever defects are sparse ($3^{3j_r+4} \leq 2^{3j_{r+1}+4}$, e.g. $2j_r + 2 \leq j_{r+1}$), $U_{r+1} + 2 \leq U_r$, unconditionally excluding all infinite sparse-defect chaotic chains in the Cantor dust! - **Complete Categorization of Cantor-Dust Families: Exact 3-Adic Valuation $v_3(U_r) = 2$, Non-Stationarity $V_{r+1} \neq V_r$ & Cumulative-Epoch Exclusion (`three_pow_ne_two_pow_add_one_of_four_le`, `cantor_dust_three_adic_valuation_and_normalized_recurrence`, `cantor_dust_multi_step_telescoping_and_epoch_descent`, `no_infinite_cumulative_sparse_epoch_cantor_dust_chain`):** Across consecutive defect steps, every U_r ($r \geq 1$) satisfies $U_r = 9V_r$ with $\gcd(V_r, 6) = 1$, reducing the transition to the coprime recurrence $2^{3j_2+4}V_2 + 1 = 3^{3j_1+4}V_1$ with ($j_2 \bmod 2 = 0 \leftrightarrow V_2 \bmod 3 = 2$) and ($j_2 \bmod 2 = 1 \leftrightarrow V_2 \bmod 3 = 1$). Reducing $3^A = 2^B + 1$ ($B \geq 4$) modulo 16 and 5 proves $V_{r+1} \neq V_r$ ($U_{r+1} \neq U_r$) for all $r \geq 1$. Furthermore, telescoping across $d \geq 1$ defects proves $2^{\sum_{t=1}^{d}(3j(r_0+t)+4)}U(r_0+d) < 3^{\sum_{t=0}^{d-1}(3j(r_0+t)+4)}U(r_0)$, so any oscillating chaotic path with infinitely many cumulative sparse epochs $[R_k, R_{k+1}]$ undergoes strict descent $U(R_{k+1}) + 2 \leq U(R_k)$ and terminates! - **Exact 2-Adic Valuation Determinism, Universal Mod-16 & Mod-128 Spacing Bijections, Zero-Branching Uniqueness & Dual $(2,3)$-Adic CRT Renormalization (`defectNormalizedQuotient_exact_two_adic_valuation_and_forward_uniqueness`, `defectNormalizedQuotient_mod_sixteen_spacing_bijection`, `defectNormalizedQuotient_orbit_unique_of_V0`, `defectNormalizedQuotient_binary_crt_parametrization`, `defectNormalizedQuotient_cooldown_mod_128_and_three_adic_lock`, `defectNormalizedQuotient_renormalized_k_recurrence`):** Because V_{r+1} is odd, $2^{3j_{r+1}+4}$ is the exact power of 2 dividing $3^{3j_r+4}V_r - 1$, so (V_r, j_r) uniquely forces (j_{r+1}, V_{r+1}). Moreover, $3^{3j_r+4}V_r \equiv 1 \pmod{16}$ restricts $V_r \bmod 16 \in \{1, 3, 9, 11\}$ and bijectively determines $j_r \bmod 4$ ($0 \leftrightarrow 1$, $1 \leftrightarrow 3$, $2 \leftrightarrow 9$, $3 \leftrightarrow 11$); hence any two trajectories in $\{0,1,2,3\}^{\mathbb{N}}$ (and in particular in the binary Cantor dust $\{1,2\}^{\mathbb{N}}$) with $V(0) = V'(0)$ are identically equal at every step (**zero branching**). When $j_{r+1} \geq 1$ (on $[280/2059, 1/2)$), $3^{3j_r+4}V_r \equiv 1 \pmod{128}$ forces $V_r \bmod 128 = 35$ ($j_r = 1$) or 25 ($j_r = 2$), eliminating 126 of 128 residue classes; combining this 2-adic lock with the 3-adic lock $V_{r+1} \bmod 3^{3j_r+4} \in \{598, 1715\} \pmod{2187}$ or $\{11533, 23585\} \pmod{59049}$ yields the exact single-parameter CRT representation $(V_r, V_{r+1}) = (2^{3j_{r+1}+11}k_r + R_0,\ 128 \cdot 3^{3j_r+4}k_r + R_1)$ with composite V_{r+1}-moduli 279,936 and 7,558,272, and the exact two-step renormalized recurrence $3^{3j_r+4}k_r - 2^{3j_{r+2}+4}k_{r+1} = C(j_r, j_{r+1}, j_{r+2})$ across all 8 triples in $\{1,2\}^3$. - **Level-2 and Level-3 CRT Renormalizations, Extinction of All $\ell_1 = 0$ Base Witnesses ($\ell_1 \geq 7$, $V_1 \geq 94{,}389{,}155$), and the General Multi-Step Homogeneous $(2,3)$-Adic Tower Law (`defectNormalizedQuotient_level2_crt_parametrization`, `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_one`, `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_two`, `defectNormalizedQuotient_multi_step_homogeneous_tower`):** Solving the 8 Level-1 renormalized equations locks $(k_1, k_2, V_1, V_2, V_3)$ across $\{1,2\}^3$ in terms of $\ell_1 = (V_1/2^{3j_2+11})/2^{3j_3+4}$, with pure 2-adic modulus $2^{3j_2+3j_3+15}$ on V_1, mixed $(2,3)$-adic modulus $2^{3j_3+11} \cdot 3^{3j_1+4}$ on V_2, and 3-adic modulus $128 \cdot 3^{3j_1+3j_2+8}$ on V_3. Chaining three transitions $(j_1, j_2, j_3, j_4) \in \{1,2\}^4$ yields the 16 Level-3 renormalized recurrences $3^{3j_1+4}\ell_1 - 2^{3j_4+4}\ell_2 = D(j_1, j_2, j_3, j_4)$, forcing $\ell_1 \pmod{2^{3j_4+4}} \in \{45, 756, 73, 910, 9, 78, 65, 864, 105, 558, 125, 412, 28, 571, 7, 980\}$ ($\ell_1 \geq 7 > 0$ in every case) so that every $\ell_1 = 0$ base witness dies at Step 4 and $V_1 \geq 94{,}389{,}155$ across all 16 4-tuples. Finally, for any spacing sequence $j : \mathbb{N} \to \mathbb{N}$ and any two solutions V, V', every origin r_0 and depth d satisfy $2^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+1+t), d)}V(r_0+d) + 3^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+t), d)}V'(r_0) = 2^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+1+t), d)}V'(r_0+d) + 3^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+t), d)}V(r_0)$. - **Base-$3^{3j_1+4}$ Carry-Digit Identity, Verified Backward 2-Adic Renormalization Tree Engine & Kernel-Certified Horizon Bounds Through Depth 10 ($V_0 \geq 7.729 \times 10^{21} > 2^{72}$ Across All 2,048 Words in $\{1,2\}^{11}$) (`renormalized_three_adic_carry_digit_identity`,**

`backward_two_adic_residue_lift_of_stepOk`, `binary_cantor_dust_min_seed_of_certifyBinaryDustDepth`, `binary_cantor_dust_certified_horizon_bounds_up_to_depth_ten`**):** Across any step $2^{3j_2+4}V_2+1=3^{3j_1+4}V_1$ with $V_1 \bmod 2^{e_2+(3j_2+4)}=R_1$ and $V_2 \bmod 2^{e_2}=R_2$, the renormalized parameters $\ell_1=V_1/2^{e_2+(3j_2+4)}$ and $\ell_2=V_2/2^{e_2}$ satisfy $\ell_2=3^{3j_1+4}\ell_1+K_2$ where $K_2=\lfloor 3^{3j_1+4}R_1/2^{e_2+(3j_2+4)}\rfloor<3^{3j_1+4}$ is the exact base-3^{3j_1+4} carry digit ($3^{3j_1+4}R_1=2^{e_2+(3j_2+4)}K_2+2^{3j_2+4}R_2+1$), and $\ell_1=0\wedge\ell_2=0 \iff 3^{3j_1+4}V_1<2^{e_2+(3j_2+4)}$. Combining `Nat.ModEq.cancel_left_of_coprime` on $\gcd(2^{e'},3^{3j_1+4})=1$ with a 6-step Hensel/Newton quadratic-doubling inverse `oddInvModTwoPow` and a recursive tree verifier `certifyBinaryDustDepth`, the Lean 4 kernel (`by decide`) certifies all 4,092 binary words from $d=1$ through $d=10$, proving $V_0\geq 590{,}121{,}613{,}202{,}017{,}059>2^{59}$ across all 512 words in $\{1,2\}^9$ ($d=8$) and $V_0\geq 7{,}729{,}594{,}727{,}432{,}150{,}074{,}275>2^{72}$ across all 2,048 words in $\{1,2\}^{11}$ ($d=10$, over $258{,}000\times$ larger than `recordHolderA0`). - **Universal $(2,3)$-Adic Quotient-Remainder Conservation Law, Multi-Step Quotient Duality, Depth-12 Binary Bounds ($V_0>1.13\times10^{27}>2^{89}$), and Ternary $\{1,2,3\}$ Tree Certification (`one_step_quotient_and_remainder_conservation`, `universal_two_three_adic_quotient_duality`, `binary_cantor_dust_certified_depth_11_and_12`, `defectNormalizedQuotient_ternary_cooldown_mod_128_sieve`, `ternary_cantor_dust_min_seed_of_certifyTernaryDustDepth`, `ternary_cantor_dust_certified_horizon_bounds_up_to_depth_six`):** For any $A\geq1, B\geq2, M\geq1$ and $BV_2+1=AV_1$, we prove $1\leq V_1 \bmod (MB)<MB$, the exact quotient conservation $V_2/(MA)=V_1/(MB)$, and the exact remainder recurrence invariance $B(V_2 \bmod (MA))+1=A(V_1 \bmod (MB))$. Iterating across d defect steps proves that for every $r_0,d\in\mathbb{N}$ and $M\geq1$, the 3-adic quotient of $V(r_0+d)$ equals the 2-adic quotient of $V(r_0)$:

$$\frac{V(r_0+d)}{M\cdot 3^{\mathrm{defectStepSum}(\mathrm{fun}\ t\Rightarrow j(r_0+t),\,d)}}\;=\;\frac{V(r_0)}{M\cdot 2^{\mathrm{defectStepSum}(\mathrm{fun}\ t\Rightarrow j(r_0+1+t),\,d)}}.$$

Pushing our binary tree verifier to $d=11$ (4,096 words) and $d=12$ (8,192 words) certifies $V(0)\geq 2{,}425{,}581{,}233{,}956{,}690{,}499{,}486{,}627>2^{81}$ ($d=11$) and $V(0)\geq 1{,}130{,}319{,}219{,}940{,}026{,}671{,}593{,}178{,}019>2^{89}$ ($d=12$); and extending the Mod-128 Cooldown Sieve to $j_r=3$ ($V_r \bmod 128=91$) certifies across all 2,187 ternary words in $\{1,2,3\}^7$ ($d=6$) that $V(0)\geq 12{,}576{,}832{,}373{,}667>2^{43}$. - **Single-Exponent Homogeneous Reduction, Universal Archimedean Floor Invariant $\lfloor 2^{E_d}V(d)/3^{S_d}\rfloor=V(0)-1$, Cyclotomic Mod-7/13/19 Automata & Crossing the 100-Bit Barrier (`defectStepSum_shift_cancellation_and_single_exponent_identity`, `defectNormalizedQuotient_archimedean_floor_invariant`, `defectNormalizedQuotient_cyclotomic_mod_7_13_19_locks`, `mahler_min_seed_of_mahlerLeafOk`, `mahler_binary_cantor_dust_min_seed_of_certifyMahlerBinaryDustDepth`, `mahler_binary_cantor_dust_certified_bounds_up_to_depth_12`):** Telescoping $E_d=S_{d+1}-(3j(0)+4)=S_d+3(j(d)-j(0))$ collapses the d-step system to the single-exponent homogeneous equation $2^{S_{d+1}}V(d)+\Delta(j,d)=2^{3j(0)+4}\cdot3^{S_d}V(0)$, where $4\Delta(j,d)+2^{S_d}\leq3^{S_d}$. This pins $2^{E_d}V(d)$ into the narrow archimedean corridor $64\cdot3^{S_d}V(0)<64\cdot2^{E_d}V(d)+3^{S_d}$ and makes $\lfloor2^{E_d}V(d)/3^{S_d}\rfloor=V(0)-1$ a global constant of motion for all $d\geq1$. Moreover, cyclotomic reduction modulo 7, 13, 19 proves the Mod-7 forward automaton ($2^{3j_2+4}\equiv2 \pmod 7$), the Mod-13 backward automaton ($3^{3j_1+4}\equiv3 \pmod{13}$), and the Mod-19 Mahler integrality conservation law ($2^{3j_r+3}V_r$) $\bmod 19=1$ ($V_r \bmod 19\in\{11,18,7\}$ for $j_r=1,2,3$). Combining the Mod-19 lock with our backward 2-adic tree via coprime cancellation lifts each leaf modulo $19\cdot2^{e_0}$ and certifies across all 8,192 words in $\{1,2\}^{13}$ ($d=12$) that $V(0)\geq 6{,}644{,}210{,}551{,}001{,}288{,}379{,}891{,}372{,}825>2^{92}$ and $9\cdot2^{3j(0)+3}V(0)>2^{101}$—crossing the **100-bit barrier** in the Lean 4 kernel. - **Advancing on the Dense Core $\{1,2\}^{\mathbb{N}}$ ($d=\infty$): Strict Doubling Growth, Repeated Defect-Subword Cancellation $2^E\mid(V(r_2)-V(r_1))$, and Unconditional Exclusion of Low-Complexity Dense-Core Trajectories (`dense_core_strict_doubling_and_monotonicity`, `dense_core_repeated_subword_divisibility_and_exp_bound`, `dense_core_repeated_subword_cutoff_and_complexity_exclusion`):** In the Dense Core $j(r)\in\{1,2\}$, $3^{3j(r)+4}\geq2187>2048=2\cdot2^{3j(r+1)+4}$ forces $V(r+1)\geq2V(r)+1>V(r)$ ($r_1<r_2\implies V(r_1)<V(r_2)$). Whenever a defect-spacing subword of length $L+1$ repeats at $r_1<r_2$ ($j(r_1+t)=j(r_2+t)$ for $0\leq t\leq L$), subtracting the homogeneous tower identities cancels the inhomogeneous accumulator completely:

$$2^E\big(V(r_2+L)-V(r_1+L)\big)\;=\;3^S\big(V(r_2)-V(r_1)\big),$$

forcing $2^E\mid(V(r_2)-V(r_1))$, $3^S\mid(V(r_2+L)-V(r_1+L))$, and $2^{7L}\leq2^E<V(r_2)$. Combining this with Section 28's archimedean upper bound $2^{E_{r_2}}V(r_2)<3^{S_{r_2}}V(0)$, the term-by-term bound $3^{S_{r_2}}\leq2^{S_{r_2}+6r_2}$, and endpoint cancellation $S_{r_2}\leq E_{r_2}+3$ proves that for $V(0)<2^W$, any repeated subword of length $L+1$ at $r_1<r_2$ must satisfy $7L<6r_2+W+3$—unconditionally excluding all Dense-Core trajectories satisfying `prefixFactorCount j (L + 1) C ≤ C` with $6C+W+3\leq7L$! - **Further Advancing the Dense Core $\{1,2\}^{\mathbb{N}}$: Right-Special 2-Adic Valuation Locks, the 7-Bit Spectral Gap, Symmetric Bispecial $(2^S,3^S)$ Resonance, Parameterized (p,q) Cutoffs & Transcendental $\frac{7\log2}{10\log(3/2)}$ Complexity Exclusion (`dense_core_all_V_odd_and_mod_four_divergence`, `dense_core_right_special_exact_two_adic_valuation_and_spectral_gap`, `dense_core_symmetric_bispecial_resonance`, `dense_core_parameterized_pq_cutoff_and_convergents`, `no_subcritical_log_threshold_dense_core`, `no_density_weighted_subcritical_log_threshold_dense_core`):** Because the $+1$ in $2^{3j(r+1)+4}V(r+1)+1=3^{3j(r)+4}V(r)$ does not depend on $j(r+1)$, matching only $K+1$ symbols ($j(r_1+t)=j(r_2+t)$ for $0\leq t\leq K$, with no match assumed at $K+1$) already cancels $+1$ at step $K\to K+1$:

$$2^{E_K}\Big(2^{3j(r_2+K+1)+4}V(r_2+K+1)-2^{3j(r_1+K+1)+4}V(r_1+K+1)\Big)\;=\;3^{S_{K+1}}\big(V(r_2)-V(r_1)\big),$$

yielding $2^{7(K+1)}\leq2^{E_K+7}<V(r_2)$ (a $+7$-bit upgrade at every length). When $j(r_1+K+1)\neq j(r_2+K+1)$ (a **right-special factor**), dividing the left difference by 128 leaves an odd integer (`8 V - V'` or `V - 8 V'`), locking $\nu_2(V(r_2)-V(r_1))=E_K+7$ (`((V(r_2) - V(r_1)) / 2^(E_K + 7)) % 2 = 1`) whereas $j(r_1+K+1)=j(r_2+K+1)$ forces $2^{E_K+14}\mid(V(r_2)-V(r_1))$—creating a **7-bit 2-adic spectral gap**:

$$j(r_1+K+1)=j(r_2+K+1)\iff2^{E_K+8}\mid\big(V(r_2)-V(r_1)\big)\iff2^{E_K+14}\mid\big(V(r_2)-V(r_1)\big).$$

For a middle match of length $K+1$ at (r_1+1,r_2+1) (a **bispecial** configuration), canceling $+1$ at both boundaries equalizes the 2-adic and 3-adic tower exponents to $S_{\mathrm{mid}}=\sum_{t=0}^{K}(3j(r_1+1+t)+4)$:

$$2^{S_{\mathrm{mid}}}\Big(2^{3j(r_2+K+2)+4}V(r_2+K+2)-2^{3j(r_1+K+2)+4}V(r_1+K+2)\Big)\;=\;3^{S_{\mathrm{mid}}}\Big(3^{3j(r_2)+4}V(r_2)-3^{3j(r_1)+4}V(r_1)\Big).$$

Finally, applying arbitrary rational power bounds $3^q \le 2^{p+q}$ ($q \ge 1$) and canceling qE_{r_2} via Section 28's shift identity proves $7qL < pS_{r_2} + q(W+3) \le 10pr_2 + q(W+3)$ for any length-L match, yielding concrete convergent cutoffs $119L < 100C + 17(W+3)$ at $(10, 17)$ and $287L < 240C + 41(W+3)$ at $(24, 41)$, proving strict suffix aperiodicity ($\forall r_1 < r_2, \exists t \le 6r_2 + W + 3, j(r_1+t) \ne j(r_2+t)$), and excluding all complexity slopes $10\kappa \log(3/2) < 7\log 2$ ($\kappa < \frac{7\log 2}{10\log(3/2)} \approx 1.196658$) and density-weighted slopes $\mu\kappa\log(3/2) < 7\log 2$ ($\kappa < \frac{7\log 2}{\mu\log(3/2)} \in [1.1967, 1.7095]$ for $\mu \in [7, 10]$). - **Top-Part & Overflow Isolation, Bottom-Part Constant of Motion, Universal Shifted-Horizon 98-Bit Corridor & Three-Occurrence Ultrametric Pigeonhole (`bottom_part_remainder_invariance_and_floor_conservation`, `dense_core_shifted_horizon_floor_and_ninety_eight_bit_corridor`, `dense_core_repeated_window_top_part_isolation_and_squeeze`, `dense_core_three_occurrence_ultrametric_pigeonhole`):** Splitting $V(r_0) = (M \cdot 2^{E_d})\ell + R_0$ and $V(r_0+d) = (M \cdot 3^{S_d})\ell + R_d$ into their **Top Part** $\ell = V(r_0)/(M \cdot 2^{E_d}) = V(r_0+d)/(M \cdot 3^{S_d})$ and **Bottom/Overflow Parts** (R_0, R_d) cancels ℓ identically from the single-exponent homogeneous equation, proving that the Top Part is 100% drift-free ($2^{E_d}(V(r_0+d) - R_d) = 3^{S_d}(V(r_0) - R_0)$) while the Bottom Part absorbs all $+1$ drift ($2^{S_{d+1}}R_d + \Delta_d = 2^{3j(r_0)+4} \cdot 3^{S_d}R_0$), satisfies $R_0 \ge 1$, and obeys its own Constant of Motion $\lfloor 2^{E_d}R_d / 3^{S_d} \rfloor = R_0 - 1$. Applying our Depth-12 tree certificates to every shifted origin $r_0 \in \mathbb{N}$ proves $V(r_0) > 2^{89}$ universally (and $V(r_0) \ge 6{,}644{,}210{,}551{,}001{,}288{,}379{,}891{,}372{,}825 > 2^{92}$ whenever $(2^{3j(0)+3}V(0)) \bmod 19 = 1$), locking the **Top 98 binary bits** of $V(r_0+d)$ to $\frac{3^{S_d}}{2^{E_d}}V(r_0)$ via $2^{98}(3^{S_d}V(r_0) - 2^{E_d}V(r_0+d)) < 3^{S_d}V(r_0)$ for all $r_0 \in \mathbb{N}$ and $d \ge 1$. Moreover, on any repeated window of length $K+1$ at $r_1 < r_2$, the Bottom Parts match identically ($V(r_1) \bmod 2^{E_K+7} = V(r_2) \bmod 2^{E_K+7}$), isolating $V(r_2) - V(r_1) = 2^{E_K+7}(\ell_2 - \ell_1)$ where m matching bits in the Top Parts ($2^m \mid (\ell_2 - \ell_1)$) tighten the cutoff to $q(7(K+1)+m) < pS_{r_2} + q(W+3)$; and whenever a length-$(K+1)$ factor appears at **three** indices $r_1 < r_2 < r_3$, binary Pigeonhole forces at least one pair across the 7-bit spectral gap ($128 \mid (\ell_b - \ell_a)$ and $2^{E_K+14} \mid (V(r_b) - V(r_a))$), gaining $+7$ bits and forcing $7q(K+2) < pS_{r_3} + q(W+3) \le 10pr_3 + q(W+3)$. - **The Infinite $(128^d, 2187^d)$ Sieve Tower, Explicit "4 in 16,384" and "8 in 2,097,152" Bijections & the Exact Prefix-Collision Biconditional (`normalizedDefectAcc_master_identity_and_infinite_sieve_tower`, `dense_core_depth2_four_in_16384_and_depth3_eight_in_2097152_sieves`, `dense_core_exact_prefix_collision_biconditional_and_mod_cutoff`):** Factoring $2^{3j(0)+4}$ out of `singleExpDefectAcc j d` defines `normalizedDefectAcc j d` ($\tilde{\Delta}(j,d)$) and establishes the factor-free master d-step identity $2^{E_d}V(d) + \tilde{\Delta}(j,d) = 3^{S_d}V(0)$. In the Dense Core, for every horizon $d = K+1 \ge 1$, this yields both the **forward "2^d in 128^d" power-of-2 sieve** $(3^{S_{K+1}}V(0)) \bmod 2^{7(K+1)} = \tilde{\Delta}(j, K+1) \bmod 2^{7(K+1)}$ (explicitly restricting $V(r) \bmod 16{,}384 \in \{931, 2851, 4889, 8601\}$ at $d=2$ and $V(r) \bmod 2{,}097{,}152 \in \{17315, 230307, 1035043, 1952547, 2020121, 1872665, 582041, 1237401\}$ at $d=3$) and the **backward "2^d in 2187^d" power-of-3 sieve** $(2^{E_{K+1}}V(K+1) + \tilde{\Delta}(j, K+1)) \bmod 3^{7(K+1)} = 0$. Finally, induction over the 7-bit spectral gap proves the **exact prefix-collision biconditional for all horizons** $K \in \mathbb{N}$:

$$(\forall t \le K,\ j(r_1+t) = j(r_2+t)) \iff 2^{E_K+7} \mid \big(V(r_2) - V(r_1)\big) \iff V(r_1) \bmod 2^{E_K+7} = V(r_2) \bmod 2^{E_K+7},$$

proving that any two steps $r_1 < r_2$ colliding modulo 2^{10K+7} must share their entire $(K+1)$-step prefix and obey $7q(K+1) < pS_{r_2} + q(W+3) \le 10pr_2 + q(W+3)$. - **Attacking the Conspiracy Barrier: Exact $\tilde{\Delta}$ Cocycle Composition, Frozen-Remainder Stabilization Beyond Bit W, and the Return-Loop Homoclinic Denominator Syzygy (`normalizedDefectAcc_cocycle_composition_law`, `dense_core_frozen_remainder_and_exact_valuation_conspiracy_law`, `dense_core_return_loop_homoclinic_denominator_syzygy`):** Across any split horizon $d_1 + m$, $\tilde{\Delta}(j, d_1+m)$ obeys the $(2,3)$-adic cocycle law $\tilde{\Delta}(j, d_1+m) = 3^{S_{\text{mid}}}\tilde{\Delta}(j, d_1) + 2^{E_{d_1}}\tilde{\Delta}(\text{shift}_{d_1}j, m)$. For any finite seed $1 \le V(0) < 2^W$, $3^{S_d}V(0) - \tilde{\Delta}(j,d) = 2^{E_d}V(d)$ has **exact 2-adic valuation** E_d at every step $d \in \mathbb{N}$ ($\neg(2^{E_d+1} \mid (3^{S_d}V(0) - \tilde{\Delta}(j,d)))$), and once $W \le 7(K+1)$, $V(0)$ has exhausted all of its binary bits: its Top Part vanishes permanently ($V(0)/2^{E_K+7} = 0$), its sieve remainder freezes at $V(0)$ itself ($V(0) \bmod 2^{E_K+7} = V(0)$), and $V(K+1) < 3^{S_{K+1}}$. Moreover, whenever a length-$(K+1)$ subword at r_1 repeats at $r_1 + d$ ($d \ge 1$), combining the shifted master equation with Top-Part isolation eliminates $V(r_1+d)$ and yields the **Exact Homoclinic Return Syzygy**:

$$\big(3^{S_{\text{ret}}} - 2^{E_{\text{ret}}}\big)V(r_1) \ = \ \tilde{\Delta}_{\text{ret}} \ + \ 2^{E_{\text{ret}}+E_K+7}(\ell_2 - \ell_1),$$

which locks $V(r_1)$ modulo the super-modulus $2^{E_{\text{ret}}+E_K+7} \ge 2^{7(d+K+1)}$ via $((3^{S_{\text{ret}}} - 2^{E_{\text{ret}}})V(r_1)) \bmod 2^{E_{\text{ret}}+E_K+7} = \tilde{\Delta}_{\text{ret}} \bmod 2^{E_{\text{ret}}+E_K+7}$ and forces the coprime homoclinic return denominator $3^{S_{\text{ret}}} - 2^{E_{\text{ret}}}$ to divide $\tilde{\Delta}_{\text{ret}} + 2^{E_{\text{ret}}+E_K+7}(\ell_2 - \ell_1)$. - **Eliminating the Seed $V(0)$ Completely: Global Origin-to-Return Syzygy & the Seed-Free Two-Return-Loop Compatibility Obstruction (`dense_core_global_origin_return_loop_syzygy`, `dense_core_seed_free_two_return_loop_syzygy_and_congruence`):** Multiplying the return-loop syzygy by $2^{E_{r_1}}$ and substituting $2^{E_{r_1}}V(r_1) = 3^{S_{r_1}}V(0) - \tilde{\Delta}(j, r_1)$ together with the cocycle law eliminates $V(r_1)$ and yields the **Global Origin-to-Return Syzygy**:

$$C_A \cdot V(0) + D_A \ = \ \tilde{\Delta}(j, r_1+d) + 2^{M_A}(\ell_2 - \ell_1),$$

where $C_A = 3^{S_{r_1}}(3^{S_{\text{ret}}} - 2^{E_{\text{ret}}})$ is odd, $D_A = 2^{E_{\text{ret}}}\tilde{\Delta}(j, r_1)$, and $M_A = E_{r_1+d} + E_K + 7 \ge 7(r_1 + d + K + 1)$, locking $V(0)$ modulo 2^{M_A} directly via $(C_A \cdot V(0) + D_A) \bmod 2^{M_A} = \tilde{\Delta}(j, r_1+d) \bmod 2^{M_A}$. Finally, across any **two** return loops $A = (r_1, d, K)$ and $B = (r_1', d', K')$ in a Dense-Core trajectory, cross-multiplying by C_B and C_A eliminates $V(0)$ identically, and reducing modulo 2^{M_A} (for $M_A \le M_B$) eliminates both Top-Part quotients Q_A, Q_B as well—leaving a **purely combinatorial, V-free and W-free 2-adic congruence on the binary sequence $j \in \{1,2\}^{\mathbb{N}}$ alone**:

$$\Big(C_B \cdot \tilde{\Delta}(j, r_1+d) + C_A \cdot D_B\Big) \bmod 2^{M_A} \ = \ \Big(C_A \cdot \tilde{\Delta}(j, r_1'+d') + C_B \cdot D_A\Big) \bmod 2^{M_A}.$$

- **Universal 3-Step Maximal Branching Loop Spawning, the Seed-Free Bit-M_A Flip Law & the Seed-Free Two-Loop 7-Bit Spectral Gap Dichotomy (`dense_core_every_three_step_window_spawns_maximal_branching_return_loop`, `dense_core_seed_free_two_loop_bit_flip_and_spectral_gap`):** By binary Pigeonhole on $\{j(r_0), j(r_0+1), j(r_0+2)\} \subseteq \{1,2\}$ and strict suffix aperiodicity, **every** 3-step window $[r_0, r_0+2]$ in any Dense-Core trajectory automatically spawns a maximal branching return loop (r_1, d, K) with $r_1 \in \{r_0, r_0+1\}$, $d \in \{1,2\}$, $K \le 6(r_1+d) + W + 3$, $(\forall t \le K, j(r_1+t) = j(r_1+d+t)) \wedge j(r_1+K+1) \ne j(r_1+d+K+1)$, and odd Top-Part jump $Q_A \bmod 2 = 1$. Pitting any return loop $A = (r_1, d, K)$ against any later return loop $B = (r_1', d', K')$ at the bits above M_A proves that when Loop A branches at $K+1$ ($j(r_1+K+1) \ne j(r_1+d+K+1)$) and $M_A < M_B$, $\text{LHS}_{A,B}$ and $\text{RHS}_{A,B}$ are **forced to disagree at bit** M_A ($(\text{LHS}_{A,B} + 2^{M_A}) \bmod 2^{M_A+1} = \text{RHS}_{A,B} \bmod 2^{M_A+1}$ and

$\mathrm{LHS}_{A,B} \bmod 2^{M_A+1} \neq \mathrm{RHS}_{A,B} \bmod 2^{M_A+1}$), whereas when Loop A stays together at $K+1$ ($j(r_1+K+1) = j(r_1+d+K+1)$) and $M_A+7 \le M_B$, they agree for 7 more bits ($\mathrm{LHS}_{A,B} \bmod 2^{M_A+7} = \mathrm{RHS}_{A,B} \bmod 2^{M_A+7}$), establishing the **Seed-Free Two-Loop 7-Bit Spectral Gap Dichotomy**:

$$j(r_1+K+1) = j(r_1+d+K+1) \iff \mathrm{LHS}_{A,B} \bmod 2^{M_A+1} = \mathrm{RHS}_{A,B} \bmod 2^{M_A+1} \iff \mathrm{LHS}_{A,B} \bmod 2^{M_A+7} = \mathrm{RHS}_{A,B} \bmod 2^{M_A+7}.$$

- Exact Constant-Run ($2059V-1$, $58025V-1$) & Alternating-Run ($129{,}009{,}091V - \{60073, 2315\}$) Valuation Locks Across Every 3-Step Window (`dense_core_constant_run_length_2059_and_58025_valuation_locks`, `dense_core_alternating_run_129009091_valuation_locks`): Evaluating Section 33's return-loop syzygy at the local periods $d=1$ and $d=2$ proves that at **every single step** r_1 of a Dense-Core trajectory, the duration of the current constant or alternating run is 100% locked to the exact 2-adic valuation of one of four explicit linear forms: **- Run of 1s ($2059 = 3^7 - 2^7$):** $\forall t \le K+1,\ j(r_1+t) = 1 \implies 2059V(r_1) = 1 + 2^{7(K+2)}(\ell_2 - \ell_1)$, with $\nu_2(2059V(r_1)-1) = 7(K+2)$ when the run ends at $j(r_1+K+2) = 2$. **- Run of 2s ($58025 = 3^{10} - 2^{10}$):** $\forall t \le K+1,\ j(r_1+t) = 2 \implies 58025V(r_1) = 1 + 2^{10(K+1)+7}(\ell_2 - \ell_1)$, with $\nu_2(58025V(r_1)-1) = 10(K+1)+7$ when the run ends at $j(r_1+K+2) = 1$. **- Alternating `1, 2, 1, ...` Run ($129{,}009{,}091 = 3^{17} - 2^{17},\ \tilde{\Delta}_{\mathrm{ret}} = 60{,}073$):** $129009091V(r_1) = 60073 + 2^{E_K+24}(\ell_2 - \ell_1)$, with $\nu_2(129009091V(r_1) - 60073) = E_K + 24$ when the alternation breaks at $j(r_1+K+1) \neq j(r_1+2+K+1)$. **- Alternating `2, 1, 2, ...` Run ($129{,}009{,}091 = 3^{17} - 2^{17},\ \tilde{\Delta}_{\mathrm{ret}} = 2{,}315$):** $129009091V(r_1) = 2315 + 2^{E_K+24}(\ell_2 - \ell_1)$, with $\nu_2(129009091V(r_1) - 2315) = E_K + 24$ when the alternation breaks at $j(r_1+K+1) \neq j(r_1+2+K+1)$. **- Exact Run-Compression & Macro-Transition Laws ($8 \cdot 2059V = 3^{7L}U_1 + 1$ and $58025V = 3^{10L}U_2 + 8$) Across All Run Lengths $L = m+1 \ge 1$ (`dense_core_one_run_compression_and_macro_transition`, `dense_core_two_run_compression_and_macro_transition`):** Multiplying the one-step recurrence by the homoclinic denominators $2059 = 3^7 - 2^7 = 29 \times 71$ and $58025 = 3^{10} - 2^{10} = 25 \times 11 \times 211$ homogenizes every interior run transition ($2^{7m}(2059V(r_1+m)-1) = 3^{7m}(2059V(r_1)-1)$ on runs of 1s; $2^{10m}(58025V(r_1+m)-1) = 3^{10m}(58025V(r_1)-1)$ on runs of 2s) and compresses **any maximal run of length $L = m+1 \ge 1$ (including singletons $L=1$)** into a single macro-step connecting the start of the run $V(r_1)$ to the start of the next run $V(r_1+L)$: **- Run of $L \ge 1$ ones switching to 2:** $\nu_2(2059V(r_1)-1) = 7L$ ($U_1 = (2059V(r_1)-1)/2^{7L}$ is odd), and

$$8 \cdot 2059 \cdot V(r_1+L) \ = \ 3^{7L} \cdot U_1 + 1,$$

forcing $(3^{7L}U_1 + 1) \bmod 8 = 0$, $(3^{7L}U_1 + 1) \bmod 2059 = 0$, and $(8 \cdot 2059V(r_1+L)) \bmod 3^{7L} = 1$. **- Run of $L \ge 1$ twos switching to 1:** $\nu_2(58025V(r_1)-1) = 10m+7 = 10L-3$ ($U_2 = (58025V(r_1)-1)/2^{10m+7}$ is odd), and

$$58025 \cdot V(r_1+L) \ = \ 3^{10L} \cdot U_2 + 8,$$

forcing $(3^{10L}U_2 + 8) \bmod 58025 = 0$ and $(58025V(r_1+L)) \bmod 3^{10L} = 8$. **- Bypassing Exponential Word Complexity (2^L) via the Universal $4(L+1)$ Anagram Pigeonhole Theorem & Partial Drift-Cancellation Cutoff (`dense_core_anagram_weight_decomposition_and_universal_pigeonhole`, `dense_core_anagram_collision_syzygy_and_drift_cancellation_cutoff`):** Although a sequence $j \in \{1,2\}^{\mathbb{N}}$ can have up to 2^L distinct subwords of length L, its step-sum $S_L(r) = 7L + 3 \cdot \mathrm{defectWindowTwoCount}(j, r, L)$ depends only on the count of 2s in $\{0, \dots, L\}$ ($L+1$ values), and by the endpoint shift identity $E_L(r) + 3j(r) = S_L(r) + 3j(r+L)$, the combined weight-and-boundary classifier $\mathrm{defectAnagramClass}(j, L, r) = 4 \cdot \mathrm{defectWindowTwoCount}(j, r, L) + 2(j(r)-1) + (j(r+L)-1)$ takes at most $4(L+1)$ values ($< 4(L+1)$). By the Finite Pigeonhole Principle, **every** Dense-Core sequence $j \in \{1,2\}^{\mathbb{N}}$ (with zero complexity assumptions!) has an anagram pair $r_1 < r_2 \le 4(L+1)$ with $j(r_1) = j(r_2)$, $j(r_1+L) = j(r_2+L)$, $S_L(r_1) = S_L(r_2) = S$, and $E_L(r_1) = E_L(r_2) = E$, satisfying:

$$2^E\big(V(r_2+L) - V(r_1+L)\big) + \tilde{\Delta}_2 \ = \ 3^S\big(V(r_2) - V(r_1)\big) + \tilde{\Delta}_1.$$

Whenever $\tilde{\Delta}_1 \bmod 2^m = \tilde{\Delta}_2 \bmod 2^m$ for $m \le E$, coprimality forces $2^m \mid (V(r_2) - V(r_1))$ and $qm < pS_{r_2} + q(W+3) \le 40p(L+1) + q(W+3)$ ($17m < 400(L+1) + 17(W+3)$ at $(p,q) = (10, 17)$). **- The $58025 - 8 \cdot 2059 = 3^7 \cdot 19$ Homoclinic Determinant Identity, Macro-Run Interface Renormalization, Three-Run Wave Elimination & Strict Well-Founded Descent $W_4 + 2 \le W_2$ Inside the Dense Core (`dense_core_one_to_two_run_interface_syzygy_and_ten_mul_valuation_lock`, `dense_core_two_to_one_run_interface_and_wave_elimination`, `dense_core_two_one_two_wave_elimination_and_strict_W_descent`, `dense_core_multi_wave_telescoping_and_epoch_W_descent`):** Eliminating the boundary state $V(r_2)$ between a run of $L_1 = m_1 + 1$ ones and a run of $L_2 = m_2 + 1$ twos via the identity $58025 - 8 \cdot 2059 = (3^{10} - 2^{10}) - 2^3(3^7 - 2^7) = 3^7(3^3 - 2^3) = 2187 \cdot 19 = 41{,}553$ proves that every post-initial run quotient is divisible by $3^7 = 2187$ ($U_k = 2187W_k$, $W_k \bmod 2 = 1$) and collapses the Dense Core to the macro-run interface system:

$$\begin{aligned} 58025 \cdot 3^{7m_1}U_1 + 19 \ &= \ 2059 \cdot 2^{10L_2}W_2 \qquad \big(\nu_2(58025 \cdot 3^{7m_1}U_1 + 19) = 10L_2\big), \\ 58025 \cdot 2^{7L_3}W_3 + 19 \ &= \ 2059 \cdot 3^{10L_2}W_2 \qquad \big(\nu_2(2059 \cdot 3^{10L_2}W_2 - 19) = 7L_3,\ 3^{10L_2} \mid (58025 \cdot 2^{7L_3}W_3 + 19)\big). \end{aligned}$$

Eliminating W_2 across a $1 \to 2 \to 1$ wave and dividing by $58025 = 3^{10} - 2^{10}$ yields $3^{10L_2+7m_1}U_1 + 19\frac{3^{10L_2} - 2^{10L_2}}{58025} = 2^{10L_2+7L_3}W_3$, while eliminating W_3 across a $2 \to 1 \to 2$ wave and dividing by $2059 = 3^7 - 2^7$ flips the $+19$ term to the 2-adic side:

$$2^{7L_3+10L_4}W_4 \ + \ 19 \cdot \left(\frac{3^{7L_3} - 2^{7L_3}}{2059}\right) \ = \ 3^{10L_2+7L_3}W_2 \implies 2^{7L_3+10L_4}W_4 \ < \ 3^{10L_2+7L_3}W_2.$$

Consequently, even though $V(r)$ strictly grows at every step of the Dense Core, **the renormalized 2-run odd quotient W undergoes strict well-founded descent $W_4 + 2 \le W_2$ inside the Dense Core** whenever $3^{10L_2+7L_3} \le 2^{7L_3+10L_4}$ (such as on the 4-step pattern `2, 1, 2, 2` where $L_2 = 1, L_3 = 1, L_4 \ge 2$ and $3^{17} < 2^{27}$). Chaining $N \ge 1$ consecutive waves telescopes all interior 2-run lengths out of the exponent difference ($2^{S_{\mathrm{wave}}(N)+10L_2(N)}W(N) < 2^{10L_2(0)}3^{S_{\mathrm{wave}}(N)}W(0)$), unconditionally excluding any Dense-Core trajectory with infinitely many wave-descent epochs! **- Same-Middle-Run Wave Cancellation, Non-Descent Run-Growth Speed Limit ($17L_4 < 27L_2 + 7L_3$) & the ($L_2 = 1, L_3 = 1$) Alternation-vs.-Descent Dichotomy (`dense_core_same_middle_run_wave_cancellation_and_divisibility`, `dense_core_non_descent_run_growth_speed_limit_and_212_dichotomy`):** Because the inhomogeneous term $19 \cdot (3^{7L_3} - 2^{7L_3})/2059$ in the $2 \to 1 \to 2$ wave

equation depends **only on the middle** 1**-run length** L_3 (and $19 \cdot (3^{10L_2} - 2^{10L_2})/58025$ in the $1 \to 2 \to 1$ wave equation depends **only on** L_2), any two waves sharing the same middle run length cancel their $+19Q$ drift identically:

$$3^{7L_3} \cdot (3^{10L_2} W_2) + 2^{7L_3} \cdot (2^{10L_4'} W_4') \;=\; 3^{7L_3} \cdot (3^{10L_2'} W_2') + 2^{7L_3} \cdot (2^{10L_4} W_4),$$

forcing $2^{7L_3} \mid (W_2' - W_2)$ when $L_2 = L_2'$ and both $2^{7L_3+10L_4} \mid (W_2' - W_2)$ and $3^{10L_2+7L_3} \mid (W_4' - W_4)$ when $(L_2, L_4) = (L_2', L_4')$. Contrapositively, whenever a wave **avoids** strict descent ($W_2 \le W_4$), every rational power bound $3^q \le 2^{p+q}$ ($q \ge 1$) forces the run-growth speed limit:

$$10qL_4 \;<\; 10(p+q)L_2 + 7pL_3 \qquad \big(17L_4 < 27L_2 + 7L_3 \text{ at } (p,q) = (10,17)\big),$$

and across $N \ge 1$ waves with $W(0) \le W(N)$, $10qL_2(N) < 10qL_2(0) + p \cdot \mathrm{waveStepSum}(L_2, L_1, N)$. Specializing $17L_4 < 27L_2 + 7L_3$ to $L_2 = 1, L_3 = 1$ ($m_2 = 0, m_3 = 0$) proves the $(L_2 = 1, L_3 = 1)$ **Alternation-vs.-Descent Dichotomy**: either $m_4 = 0$ ($L_4 = 1$, extending the alternating `2, 1, 2` run) or $W_4 + 2 \le W_2$ (strict descent of W)!

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8. Architecture of the 142 Lean 4 Files & Verification

Every theorem in this repository is machine-checked in Lean 4.19.0 (`mahler_demand_a/`) without a single `sorry`. Each core module `Foo.lean` is paired with a verification module `FooChecks.lean` that runs `#check` and `#print axioms` on every exported theorem.

Key Module Groups

- **Foundations & Odd-Block Reduction (Modules 1–36):** `MahlerDemandA.lean` through `TwoSidedSqueeze.lean` — basic Mahler step/orbit properties, companion intervals, odd-to-odd block reduction, `coprimeRemainder`, `integerSlack`, and initial two-sided squeezes.
- **1D Dynamical Collapse & Converse Reconstruction (Modules 37–52):** `SharpenedSqueezes.lean`, `AperiodicSlackCocycle.lean`, `MinSlackDynamicalSystem.lean`, `GlobalStepRateBounds.lean`, `BaseTwoDivModLock.lean`, `CeilingSaturationSynthesis.lean`, `TelescopingSeriesDuality.lean`, `ConverseCompanionReconstruction.lean`.
- **2-Adic LTE, Cauchy Convergence & Time-Independent Bounds (Modules 53–80):** `ExplicitHorizonCutoffs.lean`, `CeilingDeficitAmplification.lean`, `UniversalTwoVisitSeparation.lean`, `MonotonicStepSumSeparation.lean`, `CeilingSaturatedModTraps.lean`, `TwoAdicLiftingExponent.lean`, `FullStepSumValuation.lean`, `CoprimeRemainderAdditionLaw.lean`, `TwoAdicCauchyConvergence.lean`, `AllHistoryQuotientRecovery.lean`, `ParametricRemainderCorridor.lean`, `AperiodicBoundedBlockSqueeze.lean`, `TimeIndependentDeficiencySqueeze.lean`, `SlackDeadEndsAndWitnesses.lean`.
- **Record-Holder, Sieves, Aperiodic Exclusions & Finite-Seed Isolation (Modules 81–122):**
- `MahlerGrandSynthesis.lean`: Equivalence `validBlockChain_iff_coprime_recurrence` and 113-step survival / 114-step death of `recordHolderA0 = 29871555608238677`.
- `Mod256CompleteSieve.lean`, `CoprimeUnitRigidity.lean`, `DeterminantFactorization.lean`: Mod-256 two-block sieve, unit-sum rigidity, and rank-1 determinant collapse (Dead End 12).
- `BakerMatveevFramework.lean`: Consecutive coprimality, 3-logarithm `Real.log` squeezes, and Weil height growth (Dead End 13).
- `RidoutSubspaceFramework.lean`: Intrinsic $\gcd(\Delta_m, 6) = 1$, $2^{T(m)}$-cancellation law $2^{s(m)+1} u(m) \Delta_m < Q_m$, and Q_m^{-2} Ridout bound (Dead End 14).
- `MultiOriginSubspaceSyzygy.lean`: Additive separation of variables, shift-free `minSlack` equivalence, and 3-origin Subspace syzygy (Dead End 15).
- `RepeatedSubwordComplexity.lean`: Strict quotient aperiodicity $u(r_1) < u(r_2)$, parity-enhanced common multiplier, and linear subword-complexity cutoffs.
- `SimultaneousBaseDigitRigidity.lean`: Carryless Base-3 prefix equivalence and dual Base-2 / Base-3 remainder complementarity (Dead End 16).
- `MultiStepQuotientGCD.lean`: Exact three-way multi-step GCD equality $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)})$ and reduced-denominator exponential growth (Dead End 17).
- `MahlerSeriesHankelFramework.lean`: Power-series functional equation and two-point Padé–Hankel determinant factorization (Dead End 18).
- `ThreeAdicAndSeedGCDLocks.lean`, `CoupledFractionalPartDynamics.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `ThreeAdicValuationConservation.lean`, `Mod216TwoSidedQuotientFloor.lean`: Seed/slack GCD locks, 3-adic orbit factorization $b_{k+1} = 3^{s(k)-1} u(k)$, μ_2 interval map, disjoint Lüroth–Markov partition, Mod-2304 / Mod-1728 / Mod-216 sieves, $u(k) \neq 1$ ($k \ge 1$), and 3-adic valuation conservation.
- `InfiniteZNumberBridgeAndRationalBase.lean`: Ceiling-vs-floor odd-step proof, direct `Int.fract` finite/infinite companion equivalences, upstream `Int.fract` 113-step theorem, and rational-base $T_{3/2}$ deterministic subword exclusions.
- `SturmianAndBalancedComplexityExclusion.lean`: $5:3$ and $17:10$ weighted repetition cutoffs, Finite Pigeonhole Principle on pattern classifiers, and exclusion of aperiodic ceiling-saturated $\{3,4\}$ / $[3,5]$ and weighted-balanced ($\delta \ge 1$, slope $\le (17-\delta)/10 \le 1.6$) block chains.
- `FiniteSeedCarryAndBackwardDescent.lean`: $G_{3,6}$ false-positive cycle $(5,5) \to (4,2) \to (6,3) \to (5,5)$ vs. $8^m \mid (19a_0 + 9)$ positive-seed cutoff, arithmetic block-parameter uniqueness, and backward-lifting descent dichotomy (`a < b`, `r_a = 1/3 + (2/3)^c r_b < 13/27` for $c \ge 3$).
- `SubstitutionMahlerMethodAndGrandSynthesis.lean`: Dual-place partial-sum bounds (`0 < \Sigma_m < 1/3` vs. `-(6a_0+2)/3 \le -2/3 < 0`), closed-form uniform macro-block summation (`2^{Dm} \mid ((3^D-2^D)(3a_0+1) + 2^{s(0)} R_0)`), 3-point macro-block remainder syzygy, and `mahler_grand_122_module_synthesis`.
- `EndpointCompactnessAndBalancedExclusion.lean`: Stage 0 perpetual finite survival supremum `supCompanion a_0 = ⊔ n, lowerBound a_0 n`, strict-vs-dyadic-endpoint shift alternative (`perpetual_finite_survival_endpoint_alternative`), and positive-real forward shift (`exists_positive_seed_ge_one_of_pos_companion`); standard prefix factor complexity equivalence `prefixFactorCount s L C ≤ C ↔ ∃ r1 < r2 ≤ C, ...` (`prefixFactorCount_le_iff_exists_repeated_subword`) and global factor complexity `globalFactorComplexity s L`; parameterized (p,q) rational power bounds $3^q < 2^{p+q}$ ($qS + 1 \le pT + qW + p$); the **real logarithmic threshold selection bridge** (`exists_pq_of_lt_log_threshold` for every $\kappa \log(3/2) < \log 2$); the **asymptotic-mean ($T(n)/n \to \lambda > 0$), normalized maximum-discrepancy convergence ($E_H/H \to 0$), uniform-window-mean, and liminf-complexity exclusions** (`maxPrefixDiscrepancy_div_tendsto_zero`,

`no_infinite_asymptotic_mean_liminf_subcritical_block_chain`, `no_infinite_tendsto_mean_log_threshold_block_chain`, `no_infinite_global_complexity_log_threshold_block_chain`, `tendsto_mean_of_uniformWindowMean`, `no_infinite_uniformWindowMean_global_complexity_block_chain`); the **scale-local balance exclusion without a global limiting mean (`no_infinite_scale_local_balanced_block_chain`) and unconditional exclusion of the concrete non-stationary oscillating-epoch family `nonStationaryEpochSeq a b` (`nonStationaryEpochSeq_no_limiting_mean`, `no_infinite_nonStationaryEpochSeq_block_chain`)**; and the **derived phase-threshold factor complexity bound** $\leq L+1$ (`mechTwoLetterSeq_finset_factor_card_le`, `mechTwoLetterSeq_prefixFactorCount_le`) yielding the **assumption-free exclusion of arbitrary two-letter** $\{a,b\}$ **($3 \leq a \leq b$) mechanical/Sturmian block chains** (`unconditional_mechanical_two_letter_finite_cutoff`, `no_infinite_mechanical_two_letter_block_chain`).

- `ExactSeedResidueTreeAndStabilization.lean`: Stage 2 bidirectional parity-prefix bijection `(∀ n < N, orbit a n % 2 = orbit b n % 2) ↔ a % 2^N = b % 2^N`, constructive `seedResidue ε N < 2^N` and bit extension $r_{N+1} = r_N + \eta_N 2^N$, exact characterization of ordinary nonnegative/positive integer seeds by residue stabilization, canonical `FeasibleSeedNode` (`FeasibleSeedNode.eq_canonical`), full companion-set membership (`validCompanionUpTo_add_offset_iff_mem_ratInterval`), **denominator-cleared signed integer margin** $D_{N+1} = \min(2^{N+1}, 3D_N - 2^{N+1}(\varepsilon_N \bmod 2))$ **(`signedMargin_cast_rat`, `validCompanionUpTo_iff_signedMargin_pos`), coupled quotient-residue-margin transition (`coupled_seedQuotient_and_margin_transition`), the 5-way general Mahler conjecture equivalence (`general_mahler_conjecture_iff_positive_stabilized_branch_escape`, `mahler_conjecture_iff_leastPositiveSurvivor_unbounded`), rank-certificate escape interface (`no_perpetual_feasible_stabilized_branch_of_rank_certificate`), scalar-margin `11 / 10101` contraction vs. `100` ceiling recovery (`signedMargin_10101_neg`, `signedMargin_100_ceiling_recovery`), Lean-certified step-28 survival / step-29 failure of seed `149` and global step-29 lower bound** $m \geq 4096$ **($B(29) \geq 4096$) (`seed_149_survives_28_and_fails_29`, `positive_survivor_step29_ge_4096`, `leastPositiveSurvivor_certified_bounds`)**, single-block spike cutoff (`no_infinite_superLacunary_block_chain`), and the **well-founded odd-quotient descent ranking** $u(m+1)+2 \leq u(m)$ **and** $u(m)=1$ **mod-8 trap** (`no_infinite_rapid_growth_block_chain`) excluding all rapid-growth chains with $3 \leq s(m+1)$ and $3^{s(m)} < 2^{s(m+1)}$.
- `ThueMorseMahlerFunctionalEquation.lean`: Stage 3 formalization of the Thue–Morse sequence (`tmBit (2k) + tmBit (2k+1) = 1`), its 3-step encoded Mahler stream (`0 ≤ fractionalTail ≤ 9/19 < 1/2`, 1 odd step per 6-step aligned window), closed finite and infinite Mahler functional equations $F(z) = (1-z)F(z^2) + \frac{z}{1-z^2}$ (`tmSeries_mahler_functional_equation`), 2-adic seed Mahler bridge `64^M | (729^M (665 a_0 + 72) + 113715 * tmIntCarry 64 729 M)`, **unconditional non-eventual-periodicity** (`tmBit_not_eventually_periodic`, `tmEncodedParity_not_eventually_periodic`, `pdBit_not_eventually_periodic`, `pdEncodedParity_not_eventually_periodic`), **reusable dyadic-inflation repetition cutoff (`dyadic_inflated_repetition_cutoff`), finite seed cutoffs ($< 30 \cdot 2^k$ and $< 60W$ for `tmEncodedParity`; $< 21 \cdot 2^k$ and $< 42W$ for Period-Doubling `pdEncodedParity_cutoff_and_exclusion`), quantitative seed-residue lower bounds** $\mathrm{seedResidue}(\varepsilon, 30 \cdot 2^k) \geq 2^{2^k}$ **and bounded waiting window** $[N_0, 60N_0)$ **for a forced nonzero extension bit (`seedResidue_tmEncodedParity_ge_two_pow`, `tmEncodedParity_forced_seedBit_in_sixty_mul_window`, `seedResidue_pdEncodedParity_ge_two_pow`), and unconditional exclusion of two-letter Thue–Morse block-length chains** $s(n) = a + (b-a)t_n$ **($3 \leq a \leq b$) (`tmBlockSeq_finite_seed_cutoff`, `no_infinite_tmBlockSeq_block_chain`)**.
- `SparseDefectCocycleAndValuationLock.lean`: Stage 4 signed single-block defect identity $2^{s(k)} I(b_{k+1}) = 3^{s(k)} I(b_k) + (9 \cdot 2^{s(k)} - 8 \cdot 3^{s(k)-1})$, constant 2-adic valuation $v_2(E(c)) = 3$ for all $c \geq 4$, iterated multi-block defect cocycle and its **proved explicit `Finset.sum` expansion (`defectCocycleSum_eq_finset_sum`)**, the **short local prefix derivation (`local_prefix_first_defect_valuation_lock`)** showing that the finite prefix up to the first defect $s(j_0) \geq 4$ after j_0 blocks of 3 already forces $2^{3j_0+3} \mid (19a_0+9)$ and $2^{3j_0+4} \nmid (19a_0+9)$, multi-block valuation non-cancellation (`first_defect_valuation_lock_and_non_cancellation`), and the growing boundary invariant identity (`defectCocycle_residual_eq_growing_boundary_affInv`).
- `DubickasConfinementAndComplexityExtension.lean`: Quantitative extensions of Dubickas (2008 / 2009) on shifted carry words $s_n = qx_{n+1} - px_n$ (`dubBaseCarry_eq_fract_linear`, `dubBaseY_multi_step_carry_identity`, `dubBaseX_repeated_window_exp_and_power_cutoff`), universal two-letter alphabet $s_n \in \{\lfloor (p-q)s \rfloor, \lfloor (p-q)s \rfloor + 1\}$ (`dubBaseCarry_mem_two_letters`), general-base no-bispecial-pair property (`dubBaseCarry_no_bispecial_pair`), **finite bispecial-free binary complexity bound** $\mathrm{prefixFactorCount}(w, L, C) \leq L+1$ **(`prefixFactorCount_le_add_one_of_no_bispecial`, `dubBaseCarryNat_prefixFactorCount_le`) establishing the full general-base Dubickas (2009) Theorem 1 for all coprime** $1 < q < p < q^2$ **across all** $s \in [0, 1-1/p]$ **(`dubickas2009_general_base_explicit_and_infinite_escape`), its strictly linear-in-W bound** $N \leq 2bW + 2a + 2$ **(`dubickas2009_general_base_linear_in_W_escape`), and the sharp** $5W+14$ **escape bound at base** $3/2$ **(`shifted_third_unconditional_explicit_escape`), general-base floor-residue non-stabilization** $\lfloor \xi(p/q)^n \rfloor \bmod m$ **for all coprime** $1 < q < p < q^2$ **and** $m \geq p$ **(`general_base_floor_mod_non_stabilization`, `floor_mod_non_stabilization_of_three_le`), universal infinite-family single-cell theorem covering the open regime** $p > q^2$ **($p \geq 2q$, `general_base_non_straddling_cell_escape`), upgrade of Dubickas's Sturmian carry-word endgame ($\kappa = 1$) to the general-base critical threshold** $\kappa < \log q / \log(p/q)$ **(`exists_ab_of_lt_general_log_threshold`, `no_subcritical_general_log_threshold_dubBaseCarry`, `no_subcritical_log_threshold_dubCarry`)**, and **explicit** $O(W) = O(\log \xi)$ **finite-horizon escape bounds and infinite-horizon eventual escape** for the nearest-integer arc $\|\xi(3/2)^n\| < 1/5$ ($n \leq W+5$), the beyond-$1/3$ single window $[1/6, 13/24)$ of length $3/8 > 1/3$ at base $3/2$ ($n \leq W+15$), **the beyond-$1/4$ single window** $[1/3, 5/8)$ **of length** $7/24 > 1/4 = 1/p$ **at base** $4/3$ **($n \leq W+8$, `base_four_three_beyond_quarter_explicit_and_infinite_escape`), Dubickas's 2008 closed union** $[8/39, 18/39] \cup [21/39, 31/39]$ **of total length** $20/39 > 1/2$ ($n \leq W+8$ on the open union, $n \leq 3W+19$ on the closed union, and direct $2^m \mid (13x_r + v_r)$ divisibility on the rational 4-cycle), and **multi-interval unions of length** $7/12 \approx 0.5833$ **($n \leq W+12$), $2/3 \approx 0.6667$ ($n \leq W+14$, `union_two_thirds_explicit_and_infinite_escape`), and** $25/36 \approx 0.6944 > 2/3 > 1/2$ **($n \leq W+16$, `union_twenty_five_thirty_sixths_explicit_and_infinite_escape`) via verified backward–forward integer funnel pruning (`inRatIntervalList_terminal_of_funnelChainOk`, `inRatIntervalList_next_of_forwardStepOk`)**, plus the **Thue–Morse period-doubling Cantor cascade on** $[c, 1-c]$ **(`tmCascadeCarryAcc_mahlerProd_identity`, `tmCascadeEntry_closed_form`, `tmCascadeEntry_step_ratio_and_strictAnti`, `tmCascadeEntry_explicit_values_and_two_sevenths_cutoff`, `tmCascade_periodic_orbit_two_pow_dvd`, `cantorTwoSevenths_matches_mahlerBinaryPartition_up_to_15_and_splits_at_16`)** and the **general gap-g window ladder ($1/3 < \alpha_g < \beta_g < 1/2$) and g-rung first-defect valuation lock** $v_2((3^g - 2^g)a_0 + 3^{g-1}) = g(j_0+1)$ **(`rungAlpha_lt_rungBeta_lt_half`, `gap_ladder_short_gap_cost_and_cap`, `general_rung_first_defect_valuation_lock`)**, together with the **punctured** $[0, 1/2)$ **homoclinic catapult trap on** $[8/57, 1/2)$ **($n \leq 3W+20$, `punctured_eight_fifty_sevenths_explicit_and_infinite_escape`), the pure period-3 lock on** $[4/27, 1/2)$ **($n \leq W+8$, `punctured_four_twenty_sevenths_period_three_lock_and_escape`), the** $\varepsilon^* = 8/57$ **homoclinic phase transition**

(`homoclinicCycleMin_closed_form_and_strictMono` , `cantorPuncturedHalf_phase_transition_counts`), and the universal rung-g catapult staircase $\varepsilon_g^* = \frac{2^g}{3(3^g-2^g)} \to 0$, leak-free inductive elevator $s_{m+1} \le s_m$, self-cleaning death-band decomposition $D_{g,m}(\varepsilon)$ ($18/19 \approx 94.74\%$ of L_3 cleaned at $8/57$), 7-step catapult-into-death-band trap below $8/57$, higher-order gap-to-trigger bijection $\mathrm{Gap}_{g,m}(\varepsilon) \xrightarrow{T_g^m} [\varepsilon, b_g)$, $2^{7m} \mid (2059x_r + 280)$ leak-threading survivor lock, universal homoclinic cycle lock family $2^{(3k+4)m} \mid (D_k x_r + N_k)$, general rung-g defect cocycle identity, mutually incompatible multi-defect valuation locks, the two-island punctured-$[0,1/2)$ set $\mathcal{W}_{280/2187}$ (73.43% of $[0,1/2)$, $n \le 3W + 27$), the self-refining funnel engine and $17/24 > 70\%$ 7-interval union ($n \le W + 17$), the arbitrary K-defect simultaneous valuation lock tower, and the k-block minimum defect spacing law and $(3, 3k+4)$ Perron–Fibonacci hierarchy inside the Cantor dust below $8/57$ (`rungCatapult_ordering_and_corridor_overlap` , `universal_rung_catapult_and_upward_expansion` , `leak_free_inductive_elevator_monotone_rung_descent` , `rungDeathBand_step_and_measure_decomposition` , `catapult_into_death_band_seven_step_escape` , `rungDeathGap_maps_onto_catapult_zone` , `homoclinic_seven_cycle_in_gap_and_divisibility_lock` , `homoclinic_family_cycle_divisibility_lock` , `rungAffInv_general_defect_step_identity` , `exactTwoPowLock_incompatible_of_ne` , `general_rung_two_defect_simultaneous_lock` , `two_island_punctured_half_explicit_and_infinite_escape` , `inRatIntervalList_terminal_of_selfFunnelChainOk` , `union_seventeen_twenty_fourths_explicit_and_infinite_escape` , `rungDefectCocycle_tower_identity` , `general_rung_multi_defect_valuation_tower` , `cantor_dust_minimum_defect_spacing_law` , `punctured_eps1_forced_seven_step_macro_block` , `cantorDust_perron_fibonacci_counts` , `perronCharPoly_sign_change_and_root_squeeze` , `universal_catapult_death_band_ladder` , `punctured_eight_eighty_firsts_forbids_four_zeros_and_short_gaps` , `cantor_dust_defect_to_defect_quotient_recurrence` , `cantor_dust_mod_eight_lock_and_sparse_descent` , `no_infinite_sparse_defect_cantor_dust_chain` , `three_pow_ne_two_pow_add_one_of_four_le` , `cantor_dust_three_adic_valuation_and_normalized_recurrence` , `cantor_dust_multi_step_telescoping_and_epoch_descent` , `no_infinite_cumulative_sparse_epoch_cantor_dust_chain` , `defectNormalizedQuotient_exact_two_adic_valuation_and_forward_uniqueness` , `defectNormalizedQuotient_mod_sixteen_spacing_bijection` , `defectNormalizedQuotient_orbit_unique_of_V0` , `defectNormalizedQuotient_binary_crt_parametrization` , `defectNormalizedQuotient_cooldown_mod_128_and_three_adic_lock` , `defectNormalizedQuotient_renormalized_k_recurrence` , `defectNormalizedQuotient_level2_crt_parametrization` , `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_one` , `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_two` , `defectNormalizedQuotient_multi_step_homogeneous_tower` , `renormalized_three_adic_carry_digit_identity` , `backward_two_adic_residue_lift_of_stepOk` , `binary_cantor_dust_min_seed_of_certifyBinaryDustDepth` , `binary_cantor_dust_certified_horizon_bounds_up_to_depth_ten` , `one_step_quotient_and_remainder_conservation` , `universal_two_three_adic_quotient_duality` , `binary_cantor_dust_certified_depth_11_and_12` , `defectNormalizedQuotient_ternary_cooldown_mod_128_sieve` , `ternary_cantor_dust_min_seed_of_certifyTernaryDustDepth` , `ternary_cantor_dust_certified_horizon_bounds_up_to_depth_six` , `defectStepSum_shift_cancellation_and_single_exponent_identity` , `defectNormalizedQuotient_archimedean_floor_invariant` , `defectNormalizedQuotient_cyclotomic_mod_7_13_19_locks` , `mahler_min_seed_of_mahlerLeafOk` , `mahler_binary_cantor_dust_min_seed_of_certifyMahlerBinaryDustDepth` , `mahler_binary_cantor_dust_certified_bounds_up_to_depth_12` , `dense_core_strict_doubling_and_monotonicity` , `dense_core_repeated_subword_divisibility_and_exp_bound` , `dense_core_repeated_subword_cutoff_and_complexity_exclusion` , `dense_core_all_V_odd_and_mod_four_divergence` , `dense_core_right_special_exact_two_adic_valuation_and_spectral_gap` , `dense_core_symmetric_bispecial_resonance` , `dense_core_parameterized_pq_cutoff_and_convergents` , `no_subcritical_log_threshold_dense_core` , `no_density_weighted_subcritical_log_threshold_dense_core` , `bottom_part_remainder_invariance_and_floor_conservation` , `dense_core_shifted_horizon_floor_and_ninety_eight_bit_corridor` , `dense_core_repeated_window_top_part_isolation_and_squeeze` , `dense_core_three_occurrence_ultrametric_pigeonhole` , `normalizedDefectAcc_master_identity_and_infinite_sieve_tower` , `dense_core_depth2_four_in_16384_and_depth3_eight_in_2097152_sieves` , `dense_core_exact_prefix_collision_biconditional_and_mod_cutoff` , `normalizedDefectAcc_cocycle_composition_law` , `dense_core_frozen_remainder_and_exact_valuation_conspiracy_law` , `dense_core_return_loop_homoclinic_denominator_syzygy` , `dense_core_global_origin_return_loop_syzygy` , `dense_core_seed_free_two_return_loop_syzygy_and_congruence` , `dense_core_every_three_step_window_spawns_maximal_branching_return_loop` , `dense_core_seed_free_two_loop_bit_flip_and_spectral_gap` , `dense_core_constant_run_length_2059_and_58025_valuation_locks` , `dense_core_alternating_run_129009091_valuation_locks` , `dense_core_one_run_compression_and_macro_transition` , `dense_core_two_run_compression_and_macro_transition` , `dense_core_anagram_weight_decomposition_and_universal_pigeonhole` , `dense_core_anagram_collision_syzygy_and_drift_cancellation_cutoff` , `dense_core_one_to_two_run_interface_syzygy_and_ten_mul_valuation_lock` , `dense_core_two_to_one_run_interface_and_wave_elimination` , `dense_core_two_one_two_wave_elimination_and_strict_W_descent` , `dense_core_multi_wave_telescoping_and_epoch_W_descent` , `dense_core_same_middle_run_wave_cancellation_and_divisibility` , `dense_core_non_descent_run_growth_speed_limit_and_212_dichotomy`).

Reproducing the Verification Build

```
cd mahler_demand_a
lake build
lake env lean -DwarningAsError=true EndpointCompactnessAndBalancedExclusionChecks.lean
lake env lean -DwarningAsError=true ExactSeedResidueTreeAndStabilizationChecks.lean
lake env lean -DwarningAsError=true ThueMorseMahlerFunctionalEquationChecks.lean
lake env lean -DwarningAsError=true SparseDefectCocycleAndValuationLockChecks.lean
lake env lean -DwarningAsError=true DubickasConfinementAndComplexityExtensionChecks.lean
```

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